nature portfolio

Peer Review File

Towards Generalist Foundation Model for Radiology by Leveraging Web-scale 2D&3D Medical Data

Corresponding Author: Professor Weidi Xie

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Noteworthy Results: The paper introduces the Radiology Foundation Model (RadFM), a pioneering effort to develop a generalist medical foundation model for radiology. The authors have constructed a substantial Medical Multi-modal Dataset (MedMD) consisting of 13 million 2D images and 615,000 3D scans from publicly available datasets. Additionally, they propose an architecture that integrates text inputs with 2D or 3D medical scans to perform various radiologic tasks such as diagnosis, VQA, report generation, and rationale diagnosis. The RadFM model has demonstrated superior performance compared to some existing multi-modal 2D foundation models.

Significance to the Field: The work presents significant advancements in the field of medical AI, particularly in the development of generalist foundation models for radiology that can handle 2D and 3D data. Additionally, the utilization of longitudinal data is very important in clinical practice, and this framework enables the use of historical data. Furthermore, the paper introduces another important task, Rationale Diagnosis, which may be very useful in clinical practice. The model's ability to handle both 2D and 3D medical images, coupled with its superior performance on 2D tasks, represents significant progress in the field.

Support for Conclusions and Claims: Even though the paper introduces many important points, these aspects are not evaluated or clearly explained. First, the paper claims that the main contribution is the adaptation of the foundational model to handle both 2D and 3D data. However, 3D tasks are not detailed and evaluated in contrast to 2D ones. Also, the paper claims another important contribution is integrating different images at once to enable the utilization of longitudinal data. However, how the use of historical data affects performance is not evaluated. Since prompt selection is very important for these foundational models, the paper could include some experiments on prompt engineering, especially for report generation. The human rating section could be detailed, particularly regarding the experience and expertise of the radiologists. Additionally, since there are many public datasets for abnormality segmentation of 2D modalities, an assessment of segmentation could be included.

Flaws in Interpretation: For a comprehensive analysis, a few areas could benefit from further clarification:

(1) Details on Dataset Construction: More information on the sources and selection criteria for the datasets would provide better transparency and reproducibility.
(2) Handling of Missing Metadata: The paper mentions the lack of metadata for 3D images, which could affect the accuracy of certain tasks. A detailed discussion on how this limitation was addressed would be beneficial.
(3) Generalization Across Different Modalities: While the model performs well on multiple tasks, it would be helpful to understand how it generalizes across less common imaging modalities that were not heavily represented in the dataset.
(4) Detailed Evaluation for 3D Tasks: It is understandable that 3D data is less common, but some public 3D datasets are available and could be utilized, or some private datasets could be used. Because the main contribution of this paper is the adaptation of 3D and 2D data, this is critical.
(5) Prompt Engineering: There could be some experiments to evaluate the effects of prompt selection for captioning and diagnosis tasks.
(6) Evaluation of Historical Data: Another contribution of the paper is integrating many images as input at once. However, the utilization of longitudinal data is not evaluated.
(7) Model Architecture: It is understandable that there are no experiments on model architecture due to huge data and

computational constraints. However, there could be some experiments with a small portion of the dataset, such as encoder selection, patch size, etc.

Methodology Soundness: The model design and evaluation could be improved. Replicating 2D images to make all 2D images 3D sounds simplistic since the main contribution of this paper is creating a framework that handles both 2D and 3D. Considering that evaluation in this paper mostly focuses on 2D, replicating 2D images and making them 3D would increase computational costs unnecessarily. There are some vision encoders that can handle both 2D and 3D images, such as LongNet (Ding et al., Scaling Transformers to 1,000,000,000 Tokens, 2023) and autoregressive transformers (Hamamci et al., Text-Conditional Generation of 3D Chest CT Volumes, 2023). Also, the selection of the components for the architecture seems random. There could be some experiments with a small portion of the dataset and/or justification for the choices.

Reproducibility: The authors do not provide sufficient detail in the methods section to allow for reproduction of the work. However, they share the code and public datasets used in the study for reproducibility.

Overall, the presented work can be an important contribution to advance the field. I would recommend major revisions to address the points mentioned above.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)
The paper introduces RadFM (Radiology Foundation Model), a generalist AI model designed to perform a range of radiological tasks by integrating 2D and 3D medical images with textual data. RadFM is trained on a large, multimodal dataset (MedMD) and subsequently fine-tuned on a radiology-specific subset (RadMD). The model's performance is assessed using a new benchmark, RadBench, where it reportedly outperforms existing models in tasks such as diagnosis, visual question answering, and report generation. However, the paper acknowledges several challenges, including the limited proportion of 3D data, insufficient handling of modality-specific differences, and lack of adherence to clinical guidelines, which could impede its practical application in clinical settings.
1. Clinician Evaluation and Expertise: The evaluation of RadFM's performance was conducted by three radiologists, but the manuscript does not provide detailed information about their qualifications, years of experience, or specific areas of expertise. This omission is problematic, especially in the field of radiology and AI, where the rigor of the evaluation depends heavily on the expertise of the clinicians involved. For complex tasks, such as neuroimaging, it is essential that evaluations are conducted by specialists like neuroradiologists, and similarly, chest X-rays should be assessed by thoracic radiologists with substantial clinical experience (e.g., over 10 years). The lack of transparency in the qualifications of the evaluators raises concerns about the validity of the performance assessments. Furthermore, the manuscript does not compare RadFM's performance against state-of-the-art models such as Med-PaLM 2, which limits the ability to evaluate the claim that RadFM outperforms other models.
2. Data Composition and Modality-Specific Characteristics: The manuscript lacks clarity in explaining the composition of the dataset, particularly regarding the diverse imaging modalities included, such as CT, MRI, and mammography. Each of these modalities has distinct characteristics, including varying pixel or voxel resolutions and matrix sizes. For example, mammography involves high-resolution images with large matrix sizes, which may not be effectively managed by a unified model designed to process both 2D and 3D images without considering these differences. The proposed approach appears to oversimplify these complexities, and the manuscript does not provide sufficient transparency about how these modality-specific differences were addressed during training. This lack of detail undermines confidence in the model's generalizability across different imaging modalities.
3. Adherence to Clinical Guidelines: The quality of the radiology reports generated by RadFM is another area of concern. The reports do not appear to follow the clinical guidelines set forth by the American College of Radiology (ACR), which are crucial for ensuring that AI-generated reports are suitable for clinical use. Without adherence to these guidelines, the model's outputs cannot be considered reliable or safe for integration into clinical workflows, where precision and accuracy are paramount.
4. Innovation and Prior Work: Although the authors present RadFM as an innovative solution, similar approaches have already been published, such as Google's Med-PaLM 2. The paper does not sufficiently differentiate its contributions from these existing models, raising questions about the novelty of the proposed methods. Additionally, the manuscript lacks a thorough comparison with established models, which would have been necessary to demonstrate RadFM's unique advantages or contributions.
5. Technical and Data Transparency: The manuscript suffers from a lack of transparency in both technical implementation and data composition. Critical aspects of the training process, such as how different image resolutions and formats were handled, are not clearly described. This obscurity makes it difficult to assess the reproducibility of the study and the reliability of its results. For a model to be considered for clinical use, it is essential that every detail of its development and evaluation is meticulously documented and transparent.
6. Unified Training Structure: The authors propose a unified training structure to process both 2D and 3D images, aiming to develop a comprehensive understanding of diverse clinical images. While this approach might lead to improved versatility, it could also oversimplify the unique characteristics of different imaging modalities. This oversimplification may limit the model's ability to fully leverage the distinct features inherent to 2D and 3D data.
7. Limited 3D Data: Despite efforts to include 3D data in the training set, the dataset is predominantly composed of 2D

images. This imbalance could potentially limit the model's effectiveness in handling tasks that require a deep understanding of 3D imaging.

The authors are strongly encouraged to address these critical issues by providing detailed information about the clinicians involved in the evaluation, clarifying the data composition and handling of modality-specific characteristics, ensuring adherence to established clinical guidelines, and transparently describing the technical implementation. Addressing these concerns is essential for the work to be considered for publication.

(Remarks on code availability)
The README file lacks crucial details on the reproducible training and testing methods used in this project. It only presents the results and provides links to a website and Hugging Face repositories. This omission is concerning, as it reflects a common issue where key aspects of the methodology are not disclosed publicly, making it difficult for others to reproduce or validate the work. Transparency in the training and testing processes is essential for ensuring the credibility and reliability of the results presented. Without these details, the work cannot be fully assessed or replicated by the broader research community.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
We thank the authors for their efforts to address our comments, and to communicate their updates. This led to improvements with respect to most aspects:

1. Extended Table 2: The inclusion of data details in Extended Table 2 provides better transparency and reproducibility. This improvement is commendable.

2. Exclusion of Cases with Missing Metadata: I find the authors' explanation and handling of cases with missing metadata to be reasonable. However, as noted, this approach significantly limits the evaluation for report generation, particularly for 3D data. I strongly suggest exploring the integration of comprehensive datasets like CT-RATE (available:https://huggingface.co/datasets/ibrahimhamamci/CT-RATE) to enhance the robustness of evaluations for 3D medical imaging tasks.

3. Prompt Engineering: The additional experiments and discussions on prompt engineering appear solid and well-justified, showing that complexity in prompt design has minimal impact on their architecture.

4. Model Architecture Experiments: The authors' rationale for not testing smaller patches due to computational constraints is understandable. Similarly, their decision to retain the originally suggested architecture despite CT-ViT from GenerateCT showing slightly better performance seems pragmatic, as the improvements are marginal.

5. Model Architecture Details: Despite improvements in reproducibility through updates to the GitHub repository, the manuscript still lacks sufficient detail about the model architecture (not reproducible at all from the paper). While the code is helpful, it would be beneficial for the paper to provide a clearer explanation of the architectural choices and their implications.

6. Encoder Design: A significant limitation is the continued reliance on their old encoder, which replicates 2D images to artificially create 3D inputs. This approach appears overly simplistic, especially considering that the paper's main contribution is to establish a unified framework for processing both 2D and 3D data. Furthermore, given that the evaluation primarily focuses on 2D data, this replication unnecessarily increases computational costs without substantial benefits. It is worth noting that more advanced vision encoders are available that natively support both 2D and 3D data, providing a more efficient and performance-aligned solution. This issue is not only about performance optimization but also about the practicality of avoiding the computational overhead associated with duplicating 2D images, especially since the majority of the training and test data are inherently 2D.

Overall Assessment: The authors have addressed most of our concerns raised during the review process and demonstrated their commitment to improving the paper. Despite the noted limitations, the paper provides valuable contributions to the field of multimodal medical AI. My recommendation would be to accept the paper, with encouragement to incorporate suggestions on dataset enhancement and encoder optimization in future work.

(Remarks on code availability)
See my main comments.

Reviewer #3

(Remarks to the Author)
REVIEW FOR NATURE COMMUNICATIONS MANUSCRIPT NCOMMS-24-22213A:
"Towards Generalist Foundation Model for Radiology by Leveraging Web-scale 2D&3D Medical Data2"

This manuscript reports the results of much hard work in an area that clearly represents the future of AI in radiology. I have been asked to review the revisions of this manuscript to assess whether the revisions adequately addressed the original reviewers' concerns. The major concerns of the original reviewers are itemized below.

1. Lack of information about data sources and selection criteria. The authors mainly provide this information by citing the primary data sources and listing them in a table. However, this table should summarize what is known about the basic demographic and clinical characteristics of the patients whose images are in the data sets. I realize this information is often missing from the source publications, but some attempt needs to be made to organize and show the information that is available.

2. Insufficient description of the model. While it is commendable to make the code public, providing it without a high level explanation it not very helpful. Even with explanation, it is usually challenging to get model code to run due to its complexity. I imagine the average Nature Communications reader does not have the technical expertise to understand how the codes works simply by inspecting it. There should be some high level explanation with comparisons to prior art.

3. Unclear innovation. I am not certain how the reported work advances the field. A general radiology foundation model is certainly the future of AI in radiology, and an emphasis on data, model development, and external validation seems like a good way to frame the task. A number of commercial entities are working to develop an image-based general radiology foundation model, and there is every reason to believe they are using similar approaches. The manuscript is put forward as a "proof-of-concept", but it's unclear it has addressed any unproven concepts.

(Remarks on code availability)
I looked for a high-level explanation of the code and found none. The code also contains very little commenting.

Version 2:

Reviewer comments:

Reviewer #3

(Remarks to the Author)
I have been asked to review the revisions with respect to my comments on the previous version of this manuscript. I appreciate that the authors have done considerable work to address my comments.

1. Description of data sources. The authors have provided a nice, concise description of each data source but have not attempted to tabulate key characteristics of the sources. While it is common for data sources to omit key descriptive information, I believe it would still be useful to tabulate some basic data characteristics. Here are a few examples:

a. Original purpose of data: educational, routine clinical practice, research, etc
b. Was selection population based: all patients based on a date range or other clinical criteria (or not)
c. Was the case collection designed to represent a particular diagnosis, disease, or other clinical characteristic (or not)
d. Anatomic area(s) represented
e. Number of cases/images used
f. Country of origin
g. Was age and/or sex available in the data set

2. Description of innovation. This manuscript is apparently based on an arXiv preprint that was released 20 months ago. At this point, the information is almost historical in nature given AI's current pace of extremely rapid development. The arXiv preprint has been cited 188 times according to Google Scholar, so it has impacted the field. I will leave it to the journal editors to decide whether it is appropriate to publish this work seemingly for historical reasons. At the end of the Discussion section, the authors provide a list of 3 key concepts the work demonstrates. Even 20 months ago, I think these concepts would be viewed as fairly self-evident. The conceptual innovation still is unclear to me. I understand that the author's model is superior to others, but why? Is it simply because they incorporated more data?

3. Model description. The authors have made a reasonable effort to document their code. The high level explanatory section of the README file is a nice addition. I think the authors have done as much as can be expected to make their code reproducible.

(Remarks on code availability)

Point-point Respond to Referees: NCOMMS-24-22213-T

We thank all reviewers for their valuable comments. Here, we aim to resolve the raised concerns point-by-point, and have updated the manuscript accordingly. Please note that, our following indexes of sections, lines, figures and tables are all referred to the revised manuscript by default. All codes, datasets, and models will be released.

Reviewer 1

Q 1.1 *Details on Dataset Construction: More information on the sources and selection criteria for the datasets would provide better transparency and reproducibility.*

Reply: Thank you for the suggestions. In **Section 4.2.3**, "Training Procedure", we have included more details for selecting and filtering different datasets. Generally, in pre-training, we prioritize quantity, *i.e.*, leveraging as many available datasets as possible to maximize scale. During instruction tuning, however, we place a greater emphasis on data quality. For the ease of reading, the revised section is as follows, in **Line 598-602 and Line 609-621**:

> Generally, all the data used for model training is listed in Extended Tab. 2 with citations indicating their sources (those without citations denoting the data are contributed by this work). For pre-training, all the listed data are employed. While for domain-specific instruction tuning, we further filter out some relatively low-quality data, *i.e.*, generated data without human verification or non-radiology data, focusing more on high-quality question-answering pairs. Next, we will describe this in detail.
>
> ...
>
> **Domain-specific Instruction Tuning.** At this stage, we adopt RadMD for domain-specific instruction tuning, which contains over **3M** radiologic images, with high-quality language instructions and response. In this stage, we utilize RadMD for domain-specific instruction tuning, which includes over 3M radiological images accompanied by high-quality language instructions and responses. Notably, we filter out PMC-Inline and PMC-OA, as these datasets are not derived from real clinical scenarios. For the remaining data sources, we primarily filter out non-radiology-related content. Specifically, the filtering process targets the MPx-series, RP3D-series, and PMC-CaseReport datasets. For both MPx-series and RP3D-series, the filtering is straightforward since the original websites provide related imaging modalities for each case. For PMC-CaseReport, which is generated from the case reports subset of PMC-Inline using ChatGPT, we rely on the image captions to filter the cases. Only those with captions explicitly mentioning radiology-related terms—such as 'MRI', 'CT', 'X-ray', 'ultrasound', or 'mammography'—are retained. We acknowledge that some noisy cases may still remain in the dataset. Therefore, in our evaluation dataset, RadBench, the selected test cases undergo additional manual inspection to further ensure quality.

As mentioned above, we have listed all data sources in the **Extended Tab. 2** with citations, and for our contribution data part, we have added URL linking in the "Description" columns to indicate their raw source. The changes can also be found in **Respond Letter Table 5**.

Q 1.2 *Handling of Missing Metadata: The paper mentions the lack of metadata for 3D images, which could affect the accuracy of certain tasks. A detailed discussion on how this limitation was addressed would be beneficial.*

Reply: Thank you for your insightful suggestions regarding the handling of missing metadata for 3D images. As we mentioned in the **Discussion** section, we acknowledge that the lack of metadata, such as imaging spacing, presents a limitation for certain tasks, particularly those requiring precise measurements, such as report generation and anomaly detection. Currently, our method does not address this issue directly, and we appreciate your recommendation to elaborate on how this limitation was managed. To mitigate the impact of missing metadata on model performance, we have implemented the following measures: as indicated in **Line 428-432**.

> (iv) filter out the information about patient age or structure size, as the image spacing and patient background information are not provided. Specifically, we applied string matching techniques using Python's regular expressions to remove any sentences containing terms related to physical measurements, such as "mm", "cm", or decimal numbers (e.g., "2.5 cm"), as these are indicative of missing or incomplete metadata related to patient age, structure size, or image spacing. This step primarily addresses the problem in the report generation tasks, where such metadata would otherwise cause incorrect or unpredictable descriptions.

During the evaluation phase, we ensure that reports or cases relying on missing metadata are excluded from the test set. This helps prevent inaccuracies due to lack of image spacing or other vital information that could impact performance. We acknowledge that this decision may limit the model's ability to perform precise anomaly detection, especially in cases where accurate spatial information is required. However, as the problem of anomaly detection—particularly in medical imaging—is inherently challenging even in simpler visual-only scenarios (e.g., [16], [6]), this remains an area for future work, and we have added related discussion, in **Line 300-310**:

> *Fourth*, as the 3D images in our dataset are downloaded from the internet, some metadata is missing, for example, the imaging spacing. Such lack of precise distance measurement makes it impossible to make certain statements, such as "The tumor is 3cm large". This specification is crucial for report writing, particularly for tasks requiring precise spatial information, such as tumor size estimation, currently, we acknowledge that our model cannot be used in this way. We propose several potential solutions to address this limitation in future work, for instance, one way is to explore incorporating registration or spacing prediction models to generate the pseudo-spatial information, allowing the model to make more accurate numerical predictions related to physical measurements (e.g., tumor size). Another promising direction is to enhance the model's ability to interact with external tools or agents. For example, the model could use segmentation models alongside coding functions to calculate physical spacing directly, providing precise numeric feedback to assist in tasks like report generation or anomaly detection.

Q 1.3 *Generalization Across Different Modalities: While the model performs well on multiple tasks, it would be helpful to understand how it generalizes across less common imaging modalities that were not heavily represented in the dataset.*

Reply: Thank you for the suggestions. We have considered an extra dataset, named Lung-PET-CT-Dx [3], which is a diagnosis dataset on PET-CT series introduced in **Line 736-739**. We believe it is a less common radiology modality. The comparison results are shown in our main **Table 3** and **Respond Letter Table 3**, RadFM can also improve the final results.

Lung-PET-CT-Dx [36] consists of CT and PET-CT DICOM images of 355 lung cancer subjects. We treat it as a diagnosis dataset to further distinguish whether one patient is diagnosed with Adenocarcinoma, small cell carcinoma, large cell carcinoma, or squamous cell carcinoma. Considering its limited case number, we split it with 7:3 (train:test) to ensure enough cases for evaluation.

***Q 1.4** Detailed Evaluation for 3D Tasks: It is understandable that 3D data is less common, but some public 3D datasets are available and could be utilized, or some private datasets could be used. Because the main contribution of this paper is the adaptation of 3D and 2D data, this is critical.*

Reply: Thanks, we have added more results on 3D datasets in the main **Table 3** and **Respond Letter Table 3**. Specifically, we consider 6 more datasets for comparison, *i.e.*, MosMedData, COVID-CT, ADNI, BTM-17, Lung-PET-CT-Dx, MedDiffVQA. Among these, MosMedData, ADNI, and Lung-PET-CT-Dx all involve 3D data, introduced as follows, in **Line 711-752**:

- **MosMedData [31]** is a set of 1110 3D CT cases labeled with COVID-19 related findings, as well as without such findings. We treat it as a classification task and randomly split it with 8:2 for training and testing following [31].
- **COVID-CT [51]** is a set of 349 2D CT slices labeled with COVID-19 collected from 216 patients. We split it randomly with an 8:2 ratio for training and testing.
- **ADNI (Alzheimer's Disease Neuroimaging Initiative) [37]** is a large collection alzheimer's disease dataset with 3D brain MRI scans. We follow the setting introduced in [21] and split it randomly by 8:2 for training and testing.
- **BTM-17 (Brain-Tumor-17) [15]** is a challenge about classifying an MRI case into 17 tumor types, with 4449 real images. We adopt its official split.
- **Lung-PET-CT-Dx [36]** consists of CT and PET-CT DICOM images of 355 lung cancer subjects. We treat it as a diagnosis dataset to further distinguish whether a certain patient was diagnosed with Adenocarcinoma, small cell carcinoma, large cell carcinoma, or squamous cell carcinoma. Considering its limited case number, we split it with 7:3 (train:test) to ensure enough cases for evaluation.
- **MedDiffVQA [18]** is a large-scale dataset for difference medical VQA (involving historical comparison) in medical chest x-ray images with 700,703 pairs of question-answer. We follow its official split.

As shown by the results, RadFM can show better performance across different 3D datasets. We believe these newly added results can better support the effectiveness of our main contributions.

***Q 1.5** Prompt Engineering: There could be some experiments to evaluate the effects of prompt selection for captioning and diagnosis tasks.*

Reply: Thank you for your thoughtful comment regarding the evaluation of prompt selection in our captioning and diagnosis tasks. In response, we conducted additional experiments to investigate the effects of prompt choice, as well as other factors such as model architecture and patch size. These experiments are detailed in **Supplementary Table 3** and summarized in Table 4 (as referenced in our response letter). A more thorough discussion of these results can be found in the **"Ablation Studies"** section in **Line 102-125**, as follows:

> Beyond the data-centric ablation studies, we also conducted an additional series of experiments to demonstrate the effectiveness of our training design, as shown in Supplementary Table 3. Given the computational cost of testing these settings across the entire training dataset, we randomly sampled 10% of the RadMD dataset (using a 0.1 sample ratio) for these experiments. We used the default setup of our method—specifically, a 3D ViT model with a patch size of 32 and the prompting strategy described in the **Methods** section—and systematically varied the following factors: the image encoder architecture, the patch size, and the prompting strategy.
>
> Specifically, we vary the following factors: (i) to investigate the impact of the image encoder architecture, we experimented with more recent models, such as those proposed in [14] and [43]. (ii) we test the effects of increasing the patch size beyond 32, (iii) we experiment with more complex prompts. For example, we tested prompts with added role-play elements, such as "Assuming you are an experienced radiologist reading *modality* images, please interpret the following images and answer the related questions. This is a serious clinical case, so please be as careful and disciplined as possible. *question*," as well as incorporating varied, synonymous prompts generated by GPT-4 and verified by human reviewers.
>
> The results, summarized in Supplementary Table 3, show that modifying the image encoder architecture produced some slight gains on certain datasets, but these improvements were not statistically significant. Given that the focus of our work is not on architectural design, we chose to retain the classic 3D ViT architecture, which offers a solid baseline. When we tested the impact of increasing the patch size, we observed that larger patch sizes beyond 32 actually led to a degradation in performance. Notably, here, using smaller patch sizes might be better but they will bring unacceptable computational cost due to the increment of token number in ViT, this confirms our decision to use a patch size of 32. Lastly, variations in prompt complexity, including the use of role-playing or diverse synonymous prompts, had little to no effect on the model's performance.
>
> These additional ablation studies provide further validation for the choices we made in model architecture, patch size, and prompt design, supporting the robustness of our method.

More related to your question, we have explored adding more complex role-play instructions to our prompt strategies or increasing prompt diversity. However, neither of these adjustments appeared to influence the final performance. We believe this is because, unlike the general medical domain—where task categories are diverse and require specific, detailed prompts to guide the model's expectations—the radiology domain has more limited task categories. Consequently, prompt engineering strategies are less critical in this context since it is easy for the model to locate what role it should play.

Q 1.6 *Evaluation of Historical Data: Another contribution of the paper is integrating many images as input at once. However, the utilization of longitudinal data is not evaluated.*

Reply: Thanks. We have added extra experiments on Med-diff-VQA which is a representative historical longitudinal comparison radiology dataset, detailed introduced as in **Line 748-749**.As shown by the main **Table 3** and **Respond Letter Table 3**, for the multiple image comparison cases, our model can significantly outperform the existing SOTAs, demonstrating the effectiveness of our model for longitudinal analysis involving historical images or scans.

MedDiffVQA [18] is a large-scale dataset for difference medical VQA (involving historical comparison) in medical chest x-ray images with 700,703 pairs of question-answer. We follow its official split.

Q 1.7 *Model Architecture: It is understandable that there are no experiments on model architecture due to huge data and computational constraints. However, there could be some experiments with a small portion of the dataset, such as encoder selection, patch size, etc.*

Reply: Thank you for your insightful comment. In response to your suggestion, we have carried out a series of ablation studies to explore the impact of different model architecture choices, including encoder selection and patch size. These additional experiments are detailed in **Q1.5** and the results are presented in the supplementary materials.

- First, we tested several image encoders, including more recent models, and varied the patch size to better understand their effects on performance. While some architectural modifications, particularly the use of auto-regressive transformers, yielded slight improvements on certain datasets, these gains were not statistically significant. Given that our primary contribution is not focused on architectural innovations but rather on the overall model design and training methodology, we opted to retain the classical 3D Vision Transformer (ViT) architecture as our baseline.

- Second, we found that increasing the patch size beyond 32 led to a deterioration in model performance. This further supports our choice of a patch size of 32, as larger patch sizes resulted in diminishing returns. We also acknowledge that using smaller patch sizes could potentially improve performance, but such an approach would incur substantial computational costs due to the increased number of tokens in the ViT model.

We believe that these ablation studies address your concern and provide additional justification for our architecture choices. We hope this clarifies the rationale behind our model design and alleviates any concerns regarding the architecture.

Q 1.8 *There are some vision encoders that can handle both 2D and 3D images, such as LongNet (Ding et al., Scaling Transformers to 1,000,000,000 Tokens, 2023) and autoregressive transformers (Hamamci et al., Text-Conditional Generation of 3D Chest CT Volumes, 2023). Also, the selection of the components for the architecture seems random. There could be some experiments with a small portion of the dataset and/or justification for the choices.*

Reply: Sincerely thanks for your constructive suggestions, please check **Q1.7**. We have experimented the two models.

Q 1.9 *The authors do not provide sufficient detail in the methods section to allow for reproduction of the work. However, they share the code and public datasets used in the study for reproducibility.*

Reply: Thanks for your acknowledgment. We have further polished the README file to provide more details to reproduce our results. You can check GitHub Updating or **Respond Letter Figure 1** to see the changes. Specifically, we have added a guidance table that illustrates how different datasets are processed in our Python code and how they correspond to our released split CSV files. We have also added a **Supplementary Table 2** (**Respond Letter Table 6**) to clarify how we pre-process the data from different sources.

Reviewer 2

Q 2.1 *Clinician Evaluation and Expertise: The evaluation of RadFM's performance was conducted by three radiologists, but the manuscript does not provide detailed information about their qualifications, years of experience, or specific areas of expertise. This omission is problematic, especially in the field of radiology and AI, where the rigor of the evaluation depends heavily on the expertise of the clinicians involved. For complex tasks, such as neuroimaging, it is essential that evaluations are conducted by specialists like neuroradiologists, and similarly, chest X-rays should be assessed by thoracic radiologists with substantial clinical experience (e.g., over 10 years). The lack of transparency in the qualifications of the evaluators raises concerns about the validity of the performance assessments.*

Reply: Thank you for your valuable feedback regarding the qualifications of the radiologists involved in evaluating RadFM. We understand the importance of clinician expertise, especially in fields like radiology and AI, where the rigor of the evaluation process is crucial. In response to your concerns, we have revised the manuscript to provide additional details about the evaluation.

First, regarding the qualifications of the three radiologists: each has at least five years of clinical experience in hospital radiology departments. One radiologist is affiliated with Shanghai General Hospital, while the other two are from Shanghai Sixth People's Hospital. All three radiologists have completed their studies in Medical Imaging and Nuclear Medicine at Shanghai Jiao Tong University. Please see the updated manuscript in **Line 779-782**:

> Three radiologists were asked to rate the quality of the generated answers using a 0–5 scale. Each radiologist has five years of clinical experience in radiology departments. One is affiliated with Shanghai General Hospital, and the other two are from Shanghai Sixth People's Hospital. All three completed their studies in 'Medical Imaging and Nuclear Medicine' at Shanghai Jiao Tong University.

Second, before initiating the large-scale evaluation, we administered a preliminary exam consisting of 20 cases randomly sampled from the test set, introduced in **Line 792-798**:

> **Prior to the full evaluation**, we conducted a preliminary exam involving 20 randomly sampled test cases. This exam was designed to ensure that the radiologists understood the evaluation criteria. All three radiologists showed consistent results, with one exception: for one case, one radiologist rated the answer as 2 while the others rated it as 3. This indicates that our 5-point rating system was sufficiently clear for evaluating the model's outputs. The exam results were also reviewed by a senior radiologist with over 10 years of experience from the radiology department of Shanghai Sixth People's Hospital, further confirming the validity of the evaluation process.

Lastly, as our evaluation dataset includes peer-reviewed reports and standardized answers for reference, the evaluation does not require complex image interpretation, which would be necessary for more specialized tasks like neuroimaging or thoracic radiology. Since RadFM represents a proof-of-concept rather than a clinically validated system, the evaluators focused on assessing the textual quality of the model's outputs rather than performing detailed image-based diagnostics. Most model outputs were relatively poor, as indicated by average scores of around 2 (i.e., "The content holds some reference value, but its correctness is subpar"). Thus, the evaluators primarily compared the generated text to the reference answers, drawing on their clinical experience to assess semantic accuracy rather than interpreting medical images.

In summary, we believe that the evaluation is appropriate given the current stage of generative models in medical imaging. The radiologists involved are well-qualified to perform the evaluation task, and the process itself was straightforward, as all existing models are far from achieving clinical-level precision. These revisions should address your concerns regarding the expertise of the evaluators.

Q 2.2 *Furthermore, the manuscript does not compare RadFM's performance against state-of-the-art models such as Med-PaLM 2, which limits the ability to evaluate the claim that RadFM outperforms other models.*

Reply: Thank you for your suggestion. We understand the importance of comparing RadFM against state-of-the-art models, and we appreciate your concern. However, we would like to clarify a few points:

- **Differences Against Med-PaLM Series.** Med-PaLM 2 is a language-only model and, as such, is not directly comparable to RadFM, which is a vision-language model designed to perform tasks such as MedVQA and report generation—tasks that require integration of both image and text modalities. We assume you are referring to Med-PaLM [41], which is a model developed by Google based on the PaLM architecture. However, Med-PaLM is a closed-source model, and as such, it is not accessible for direct comparison.
- **More Comparison Baselines.** In response to your suggestion, we have included a new open-source baseline, LLaVA-Med [25], to more effectively demonstrate the performance of our model. As shown by **Respond Letter Table 1** and **Respond Letter Table 2**, or, main **Table 1** and **Table 2**, RadFM significantly outperforms LLaVA-Med on both public benchmarks and our custom benchmark, RadBench, showcasing the superiority of our method.
- **Inspiring Recent Works.** We would like to highlight that more recent medical generative vision-language models, such as BiomedGPT[48], CheXagent[11], Huatuogpt-vision[9], and MedDR[17], have been inspired by our work and have cited RadFM. Since these models were developed after RadFM, we do not consider them as fair baselines for direct comparison.

We hope this clarifies our reasoning for not including Med-PaLM 2 in the comparison and demonstrates that RadFM outperforms the current open-source alternatives.

Q 2.3 *Data Composition and Modality-Specific Characteristics: The manuscript lacks clarity in explaining the composition of the dataset, particularly regarding the diverse imaging modalities included, such as CT, MRI, and mammography. Each of these modalities has distinct characteristics, including varying pixel or voxel resolutions and matrix sizes. For example, mammography involves high-resolution images with large matrix sizes, which may not be effectively managed by a unified model designed to process both 2D and 3D images without considering these differences. The proposed approach appears to oversimplify these complexities, and the manuscript does not provide sufficient transparency about how these modality-specific differences were addressed during training. This lack of detail undermines confidence in the model's generalizability across different imaging modalities.*

Reply: Thank you for your thoughtful comment. We understand the importance of addressing modality-specific characteristics, especially when dealing with diverse imaging modalities such as CT, MRI, and mammography. To clarify, we have added additional details in the manuscript and **Supplementary Table 2** (referenced in **Respond Letter Table 6**) to explain the resolution changes and pre-processing steps applied to the different datasets.

First, we want to emphasize that the decision to use lower-resolution images in our experiments was primarily driven by computational cost, not limitations of the model itself. The 3D ViT model can process images of arbitrary resolution by working with image patch tokens, allowing it to handle both 2D and 3D images with varying pixel and voxel resolutions. However, due to the computational resources available, we downsampled images to ensure that the experiments were feasible. If additional resources (e.g., access to large GPU clusters such as those used for LLaMA) were available, we could conduct experiments at higher resolutions. As this is a proof-of-concept study, we focused on initiating the development of a medical multimodal foundation model and evaluated our model at resolutions that are consistent with previous works [41, 30, 25]. We acknowledge that scaling up the model and data to handle higher resolutions and larger datasets will be necessary for real-world clinical applications. We discuss this in the limitations section (see **Line 283-290**), as follows:

> *First*, the capacity to generate meaningful and accurate long sentences remains underdeveloped, causing the foundation models still far from clinically useful. As demonstrated in Extended Tab. 1, for rationale diagnosis and report generation, the quantitative results surpass previous works but are still far from practical satisfactory. In human rating, similar results are also observed. As shown in Fig. 4, none of the model gets over score 3 that represents moderately accurate, showing that there is still long way to go for developing generalist medical foundation models. For real clinical usage, we suggest future improvements, including scaling model sizes, increasing image resolutions, and expanding clinical data, as outlined by the scaling laws in [20].

Second, to better demonstrate the model's performance across different imaging modalities, we have added results from six additional datasets: MosMedData, COVID-CT, ADNI, BTM-17, Lung-PET-CT-Dx, and MedDiffVQA (see Respond Letter Table 3 and main Table 3). These datasets include CT, MRI, X-ray, and PET images, covering both 2D slices and 3D scans. Our results show that the model performs significantly well across these modalities, demonstrating robustness and adaptability, even with rare modalities like PET. This expanded evaluation provides a clearer understanding of the model's generalizability across different imaging types. The detailed introduction of newly added datasets is as follows, in **Line 711-752**:

> - **MosMedData [31]** is a set of 1110 3D CT cases labeled with COVID-19 related findings, as well as without such findings. We view it as a classification task and split it randomly with 8:2 for training and testing following [31].
> - **COVID-CT [51]** is a set of 349 2D CT slices labeled with COVID-19 collected from 216 patients. We split it randomly with an 8:2 ratio for training and testing.
> - **ADNI (Alzheimer's Disease Neuroimaging Initiative) [37]** is a large collection alzheimer's disease dataset with 3D brain MRI scans. We follow the setting introduced in [21] and split it randomly by 8:2 for training and testing.

- **BTM-17 (Brain-Tumor-17)** [15] is a challenge about classifying an MRI case into 17 tumor types, with 4449 real images. We adopt its official split.
- **Lung-PET-CT-Dx** [36] consists of CT and PET-CT DICOM images of 355 lung cancer subjects. We view it as a diagnosis dataset to further distinguish whether a certain Patient was diagnosed with Adenocarcinoma, small cell carcinoma, large cell carcinoma, or squamous cell carcinoma. Considering its limited case number, we split it with 7:3 (train:test) to ensure enough cases for evaluation.
- **MedDiffVQA** [18] is a large-scale dataset for difference medical VQA (involving historical comparison) in medical chest x-ray images with 700,703 pairs of question-answer. We follow its official split.

Q 2.4 *Adherence to Clinical Guidelines: The quality of the radiology reports generated by RadFM is another area of concern. The reports do not appear to follow the clinical guidelines set forth by the American College of Radiology (ACR), which are crucial for ensuring that AI-generated reports are suitable for clinical use. Without adherence to these guidelines, the model's outputs cannot be considered reliable or safe for integration into clinical workflows, where precision and accuracy are paramount.*

Reply: Thank you for your thoughtful comments. We appreciate your concerns regarding adherence to clinical guidelines, particularly the American College of Radiology (ACR) standards, in the radiology reports generated by RadFM.

We would like to emphasize that this work is primarily a **research-oriented proof-of-concept** aimed at exploring the feasibility of developing generative foundation models for radiology report generation, rather than an attempt to immediately integrate these models into clinical workflows. Our main goal is to lay the groundwork for future advancements in the field, demonstrating the potential of multimodal AI models in radiology, with a focus on processing both 3D and 2D images alongside textual data.

1. Current Focus of RadFM: We fully acknowledge that the quality of the reports generated by RadFM does not yet meet the rigorous clinical standards required for direct clinical application, including the adherence to specific guidelines like those from the ACR. This is an ongoing challenge in the AI research community, as reflected in human evaluation results showing that most models, including RadFM, still struggle to produce clinically satisfactory outputs. Specifically, in our human ratings, most models achieved an average score of 2 ("The content holds some reference value, yet its correctness is subpar"), highlighting the gap between the current state of AI-generated reports and the quality needed for clinical usage.

2. Recognizing Limitations and Next Steps: As we discuss in the Discussion section (see Line [number needed]), our paper primarily focuses on laying the foundations for medical generative vision-language models. Improving the accuracy and adherence to clinical guidelines, such as those of the ACR, is a significant challenge that extends beyond the scope of this proof-of-concept. Addressing these concerns and ensuring the clinical usability of these models will require more advanced engineering, dataset refinement, and rigorous evaluation against clinical standards, which is part of the ongoing development work. In our more recent work, AutoRG-Brain [24], we have taken steps toward addressing these challenges and combining AI models with clinical workflows.

3. Clarification of Scope in the Manuscript: We have added a clarification in the Discussion section to better highlight the research-oriented focus of this study and to set the proper context for our contributions. We have added this limitation clarification in the final "Discussion" section to better state our position in **Line 344-352**, as:

> In conclusion, this paper serves as a proof-of-concept initialization for building medical generative vision-language foundation models. We present a comprehensive set of model-building processes, including data collection, problem formulation, model design, training, and evaluation. While our model significantly outperforms existing open-source multimodal foundation models, we acknowledge that it does not yet meet clinical criteria. Nevertheless, we believe this work provides a starting point for future research and development toward more generalist medical AI models.

Furthermore, we emphasize in the **Abstract** that the focus of this work is on the **proof-of-concept** nature of our model, rather than immediate clinical application.

Q 2.5 *Innovation and Prior Work: Although the authors present RadFM as an innovative solution, similar approaches have already been published, such as Google's Med-PaLM 2. The paper does not sufficiently differentiate its contributions from these existing models, raising questions about the novelty of the proposed methods. Additionally, the manuscript lacks a thorough comparison with established models, which would have been necessary to demonstrate RadFM's unique advantages or contributions.*

Reply: Thank you for your valuable feedback. We appreciate your concerns regarding the novelty of our approach, especially in relation to existing models such as Med-PaLM 2, and the need for a more thorough comparison with established models. We would like to clarify the following points:

1. Distinguishing RadFM from Med-PaLM 2: While we acknowledge the significance of Med-PaLM 2 in the field of medical language models, we must emphasize that it is a language-only model. In contrast, RadFM is a multimodal vision-language model designed to process both medical images (including 2D and 3D images) and associated textual data, such as radiology reports. This fundamental difference in model architecture means that RadFM is capable of handling a broader range of clinical tasks, such as interpreting images alongside text and generating comprehensive radiology reports, which Med-PaLM 2 cannot achieve due to its lack of image processing capabilities.

2. Contribution and Novelty in RadFM: We also recognize that recent works have built on similar concepts, and we are glad to see that RadFM's approach has influenced several subsequent models, including BiomedGPT[48], CheXagent[11], Huatuogpt-vision[9], and MedDR[17], all of which cite our work. These developments highlight the foundational nature of RadFM, and as such, we do not consider them as direct competitors for comparison in this manuscript.

3. Comparison with Other Models: To ensure a fair comparison and better demonstrate the performance of RadFM, we have added a contemporary open-source baseline, LLaVA-Med [25], to the manuscript. Our results show that, despite the updates made to LLaVA-Med after our model's release, RadFM consistently outperforms LLaVA-Med on both public benchmarks and our own RadBench. This comparison is included in **Table 1**, **Table 2**, and further illustrated in the **Respond Letter Table 1** and **Respond Letter Table 2**. This addition reinforces the effectiveness and superiority of RadFM compared to the existing state-of-the-art models.

4. Related Work Section Revision: We have revised the Related Work section to better contextualize RadFM within the landscape of existing research and highlight its unique contributions. The updated section (see **Line 315-343**) is as follows:

Related works. With the success of generative language foundation models such as GPT-4 [35] and PaLM-2 [4], there has been a surge of interest in multi-modal foundation models. Significant strides have been made in the realm of natural scenery, as evidenced by BLIP-2 [26] and Flamingo [1]. Though in the context of the medical language-only models, great steps have been made, like Med-PALM series [40, 39], PMC-LLaMA [47], Meditron-70b [10], the development of multimodal medical artificial intelligence is still in its nascent stages [29]. The relevant research can be bifurcated into two primary areas, namely, dataset construction and model training.

- **Dataset Construction.** Contrary to the natural scenery domain, which boasts numerous large-scale multi-modal datasets such as MMC4 [52], Visual Genome [22], and LION-5B [38], the medical domain is somewhat lacking. The most widely utilized medical multi-modal dataset is MIMIC-CXR [19], which only contains chest X-ray images with caption reports and its quantity (224K) is relatively small. In PMC-OA [27], the authors have compiled a dataset containing 1.6M image-caption pairs. Although it encompasses various image modalities, many 3D medical scans are presented as 2D slices since the images are extracted from papers. There are also some medical VQA datasets, such as VQA-RAD [23], SLAKE [28], and PMC-VQA [50], but they are also limited to 2D images. In Med-Flamingo [30], they have collected a dataset, MTB, consisting of approximately 0.8M images interleaved with texts while it is not open-source. Consequently, due to the limitations on data availability, existing medical foundation models have concentrated on a narrow range of data modalities. For example, LLaVA-Med [25] and MedVInT [50] utilize image captions in PubMed Central, which exists a significant domain gap between real-world clinical data. In Med-PaLM M [41], the authors amalgamate existing medical images or multi-modal datasets, but the majority of images are X-rays, which are not sufficiently accurate for clinical practice.
- **Model Training.** To date, several works have focused on building medical foundation models, yet most of these works are limited to support for 2D images [25, 50, 41, 30]. An ideal foundation model should exhibit a comprehensive set of capabilities: it should support both 2D and 3D image inputs, multi-image input per case, and images interleaved with text inputs. Currently, there exists no model that can simultaneously support this series of heterogeneous input forms. This paper aims to address these gaps, aligning more closely with clinical practice and we are very glad to see many current works [48, 11, 9, 17] have bee inspired based on our efforts.

5. Conclusion: We hope these revisions clarify the novelty of RadFM, highlight its unique multimodal capabilities, and provide a clear distinction from existing models such as Med-PaLM 2. We also emphasize that RadFM is not just an incremental model but an important step toward bridging the gap between generative AI and clinical medical applications.

Q 2.6 *Technical and Data Transparency: The manuscript suffers from a lack of transparency in both technical implementation and data composition. Critical aspects of the training process, such as how different image resolutions and formats were handled, are not clearly described. This obscurity makes it difficult to assess the reproducibility of the study and the reliability of its results. For a model to be considered for clinical use, it is essential that every detail of its development and evaluation is meticulously documented and transparent.*

Reply: Thank you for your thoughtful comments. We fully understand the importance of technical and data transparency, especially in the context of reproducibility and clinical applicability. Please check **Q2.3** to see how we pre-process different images from various modalities. Also please check **Q2.9** to see the updated GitHub, that will better guide followers to reproduce our work.

Q 2.7 *Unified Training Structure: The authors propose a unified training structure to process both 2D and 3D images, aiming to develop a comprehensive understanding of diverse clinical images. While this approach might lead to improved versatility, it could also oversimplify the unique characteristics of different imaging modalities. This oversimplification may limit the model's ability to fully leverage the distinct features inherent to 2D and 3D data.*

Reply: Thank you for your insightful comment. We sincerely understand your concerns regarding the unified training structure and the potential risk of oversimplifying the unique characteristics of different imaging modalities. In response, we have conducted additional ablation studies to further evaluate the effectiveness of our training design and demonstrate how our approach handles both 2D and 3D images. These studies provide evidence that the unified structure is indeed effective, even though it may seem simple at first glance.

1. Ablation Studies on Training Structure: To address your concern, we have performed further ablation studies on a small subset of our datasets. These studies focus on several key aspects, including image encoder architectures, patch sizes, and prompt strategies. The results are shown in Supplementary Table 3 and Respond Letter Table 4.

- We experimented with more recent unified image encoder architectures, as recommended by Reviewer 1, and compared them against our default 3D ViT architecture. The results indicate that while more complex architectures can provide slight gains in some datasets, these improvements are not statistically significant.

- We also explored larger patch sizes beyond 32 and found that increasing the patch size actually led to a performance drop, reinforcing our choice of a patch size of 32. Using smaller patch sizes could potentially improve performance, but would also incur unacceptable computational costs due to the increase in token numbers in the ViT.

- Additionally, we tested more complex prompt strategies, including role-play prompts and diverse rewritten prompts generated by GPT-4. However, these prompt variations did not yield substantial improvements, suggesting that the current prompt design is already effective for our tasks.

2. Discussion in the Manuscript: To clarify the results and further emphasize the findings of these ablation studies, we have updated the **Ablation Studies** section in the manuscript. Please check **Q1.5** for more details.

3. Reaffirming the Main Contribution: We would like to emphasize that the primary contribution of our paper is not focused on the architectural design, but rather on developing a unified generative training pipeline capable of handling a wide variety of radiology tasks across different modalities. This unified pipeline is designed to process both 2D and 3D images in a consistent manner, which is more aligned with real-world clinical use, where data from different modalities are often combined. While architectural improvements may further optimize the model, the key innovation lies in the integration of multiple tasks within a single framework.

We hope these additions and clarifications address your concerns. The ablation studies provide a thorough evaluation of our training design, and we believe the results demonstrate that our unified approach is both effective and efficient in handling the challenges posed by different imaging modalities. Thank you again for your constructive feedback.

***Q 2.8** Limited 3D Data: Despite efforts to include 3D data in the training set, the dataset is predominantly composed of 2D images. This imbalance could potentially limit the model's effectiveness in handling tasks that require a deep understanding of 3D imaging.*

Reply: Thank you for your valuable feedback. We appreciate your point about the imbalance in the dataset, with a predominance of 2D images over 3D data. We fully acknowledge this limitation and agree that more 3D data would be beneficial for tasks requiring deep understanding of 3D imaging. However, we would like to emphasize a few important points in response:

1. Comparison with Related Work: While our dataset is predominantly composed of 2D images, it is important to note that many of the related works in the field also primarily rely on 2D images. For example:

- LLaVA-Med[25] and MedVInT[50] both primarily use 2D images extracted from medical papers.
- Med-PaLM M [41], despite considering the broader biomedical domain, also uses 2D image slices exclusively.
- Many recent works inspired by our own research, such as BiomedGPT[48], CheXagent[11], Huatuogpt-vision[9], and MedDR[17], also predominantly utilize 2D slices.

Thus, despite the limitations of our dataset in terms of 3D data, we believe that the inclusion of 3D scans in our dataset is still significant compared to other existing works, where 3D data is often not considered at all. Our approach represents an early attempt to address the gap in 3D medical data and push the development of multimodal foundation models for clinical use.

2. Improving 3D Imaging Performance: To further demonstrate the model's ability to handle 3D imaging, we have included additional results from various 3D medical datasets, such as CT, MRI, and PET scans. These results show that, despite the dataset's predominance of 2D images, our model still demonstrates significant improvements in performance across different modalities, including 3D images. We believe this is a testament to the robustness and versatility of our model.

3. Importance of the Initial Step: As we discussed in **Line 344-352** of main texts and **Q2.4**, this work serves as an initial step toward building medical multimodal foundation models that can handle both 2D and 3D data. We strongly believe that this approach is crucial for aligning models with the real demands of clinical practice, where data from diverse imaging modalities are often interleaved. Our focus on combining 2D and 3D data is intended to lay the groundwork for future improvements in handling 3D medical imaging in a more comprehensive manner.

4. Future Directions: We recognize that there is still much to be done in terms of increasing the volume and diversity of 3D data used for training. Future work will involve scaling up the 3D dataset and refining the model to better capture the nuances of 3D medical images. We will continue to explore how to effectively incorporate these 3D modalities into the model's training pipeline, as we believe that 3D data will become increasingly important in the development of clinical AI systems.

***Q 2.9** The README file lacks crucial details on the reproducible training and testing methods used in this project. It only presents the results and provides links to a website and Hugging Face* *repositories. This omission is concerning, as it reflects a common issue where key aspects of the methodology are not disclosed publicly, making it difficult for others to reproduce or validate the work.*

Reply: Thank you for your valuable feedback regarding the README file. In response to your concern, we have updated the README file to include more comprehensive details on how to reproduce the experiments and utilize the datasets used in our project.

1. Reproducibility: We have added a detailed guiding table that outlines the processing steps for each dataset used in our experiments. This table provides clear instructions on how each dataset is preprocessed, including links to the corresponding split CSV files, and how to integrate these with our Python code. This should help users understand the specific steps involved in data preprocessing and enable them to reproduce our results accurately.

2. Updated README File: You can view the updated README file with these improvements via this GitHub Link. Additionally, you can refer to **Respond Letter Figure 1** for a snapshot of the updated section that provides clarity on the dataset processing and code integration.

Respond Letter Table 1: Comparison of our proposed RadFM with other foundation models on 9 existing datasets together with ablation Studies. We adopt a few-shot prompting setting for flamingo-liked models while we adopt a zero-shot instruction prompting strategy for MedVInT, LLaVA-Med, and RadFM. "W/o Ins-tuning" denotes training without the domain-specific instruction tuning and "w/o Our Data" denotes training without all our newly collected data, *i.e.*, using the combination of existing datasets only. ACC, F1, BLEU, ROUGE, UMLS_Precision, UMLS_Recall and BERT-Sim are reported based on task types, and the metrics refer to the average score on all test samples. Numbers within parentheses indicate 95% CI. Percentage (%) signs have been omitted in the table.

Dataset	Metric	OpenFlamingo (Few-shot)	MedVInT	LLaVA-Med	MedFlamingo (Few-shot)	RadFM (w/o Ins-tuning)	RadFM (w/o Our Data)	RadFM
Disease Diagnosis								
VinDr-Mammo	ACC	49.92 (48.20, 51.65)	50.06 (48.52, 51.59)	50.27 (49.20, 51.52)	49.80 (48.33, 51.42)	49.80 (48.15,51.53)	55.35 (54.35,559.47)	**59.96 (58.41, 61.59)**
	F1	57.01 (54.64, 60.08)	**66.56 (65.2, 67.93)**	56.48 (55.45, 57.73)	64.92 (63.52, 66.32)	60.32 (58.25,62.31)	60.57 (58.75,62.58)	62.11 (60.09, 63.75)
VinDr-SpineXr	ACC	50.33 (47.13, 53.53)	49.93 (46.99, 52.86)	49.85 (47.95, 52.23)	49.61 (46.05, 53.16)	52.19 (49.18,55.17)	64.43 (61.76,67.22)	**68.82 (65.92, 71.47)**
	F1	31.79 (26.99, 36.58)	62.32 (59.38, 65.25)	54.83 (51.88, 57.45)	63.23 (59.74, 66.74)	34.19 (30.17,38.09)	65.86 (62.62,68.81)	**67.69 (64.5, 70.98)**
VinDr-PCXR	ACC	49.85 (45.40, 54.31)	50.29 (45.88, 54.69)	49.62 (45.79, 53.64)	49.37 (44.44, 54.31)	50.12 (45.21, 54.60)	51.82 (46.46, 57.09)	**56.32 (51.82, 61.21)**
	F1	41.44 (33.77, 49.10)	66.29 (62.36, 70.23)	47.81 (42.33, 53.42)	**66.94 (62.57, 71.32)**	43.33 (40.37, 40.88)	49.14 (43.66, 56.18)	37.53 (28.88, 43.67)
CXR-Mix	ACC	50.63 (50.07, 51.03)	49.2 (48.53, 49.88)	53.26 (52.72, 53.91)	50.00 (49.50, 50.51)	77.71 (77.25, 77.95)	78.63 (78.51, 79.10)	**83.62 (83.23, 83.97)**
	F1	24.83 (24.11, 25.54)	67.22 (66.62, 67.82)	22.63 (22.70, 24.53)	66.11 (65.72, 66.61)	74.42 (73.98, 75.01)	78.35 (77.85, 78.93)	**82.99 (82.58, 83.49)**
RadChest-CT	ACC	50.93 (49.13, 52.72)	50.07 (47.68, 52.45)	51.09 (50.05, 52.63)	50.39 (48.34, 52.43)	51.97 (50.05, 53.31)	69.72 (67.44, 71.53)	**72.95 (71.06, 74.78)**
	F1	43.49 (41.18, 45.99)	66.57 (64.45, 68.69)	44.42 (42.00, 46.55)	63.31 (61.39, 65.23)	38.67 (36.37, 41.46)	67.84 (65.64, 70.11)	**71.86 (69.42, 83.49)**
Medical VQA								
PMC-VQA	BLEU	11.10 (8.93, 13.41)	**23.73 (21.03, 26.73)**	13.66 (11.68, 15.52)	11.03 (9.27, 13.49)	5.23 (3.23, 8.84)	14.01 (10.92, 17.25)	17.99 (14.80, 20.83)
	ROUGE	13.03 (10.63, 15.46)	**27.24 (24.04, 30.91)**	18.14 (16.46, 20.20)	13.06 (10.93, 15.66)	5.82 (2.03, 10.09)	14.23 (11.20, 17.66)	19.43 (16.56, 23.55)
	UMLS_Precision	7.60 (5.41, 10.83)	19.64 (16.2, 23.59)	16.38 (12.67, 20.25)	6.45 (4.05, 8.97)	18.63 (14.84, 20.76)	13.24 (9.90, 17.02)	**20.74 (17.39, 24.71)**
	UMLS_Recall	7.56 (5.40, 10.51)	**18.88 (15.51, 22.68)**	13.34 (10.59, 16.07)	6.10 (4.04, 8.97)	15.03 (12.07, 18.34)	12.94 (9.39, 15.86)	14.14 (11.19, 17.37)
	BERT-Sim	52.08 (50.43, 54.07)	57.81 (55.49, 59.76)	42.46 (41.50, 43.44)	51.37 (49.57, 53.01)	47.85 (44.20, 49.37)	57.57 (55.85, 60.19)	**63.85 (62.04, 65.94)**
VQA-RAD	BLEU	33.98 (26.75, 41.85)	35.1 (28.44, 41.55)	31.55 (24.89, 38.35)	35.97 (29.14, 45.45)	22.03 (15.67, 30.38)	43.98 (36.58, 50.51)	**52.24 (44.97, 59.43)**
	ROUGE	35.26 (28.21, 43.91)	39.2 (31.36, 46.33)	37.47 (30.83, 44.47)	38.64 (31.42, 48.23)	22.67 (14.92, 28.57)	44.70 (38.35, 50.81)	**52.74 (45.39, 61.05)**
	UMLS_Precision	14.72 (6.86, 24.22)	16.46 (7.83, 25.93)	13.30 (12.14, 14.50)	18.70 (8.76, 29.61)	60.30 (50.88, 67.07)	61.52 (53.65, 69.51)	**62.12 (54.01, 71.12)**
	UMLS_Recall	14.52 (7.63, 23.33)	15.94 (7.72, 25.48)	12.16 (10.09, 13.93)	17.46 (8.76, 27.85)	39.43 (32.59, 47.12)	41.14 (34.49, 48.76)	**42.82 (32.31, 51.54)**
	BERT-Sim	71.49 (67.63, 74.96)	71.39 (66.94, 75.46)	68.28 (64.07, 72.00)	73.40 (69.62, 77.32)	58.88 (56.74, 61.08)	80.64 (77.55, 83.89)	**81.52 (77.41, 85.17)**
SLAKE	BLEU	27.16 (22.01, 32.56)	24.81 (20.23, 30.52)	21.43 (17.07, 25.35)	23.62 (18.06, 28.26)	24.39 (15.81, 30.74)	67.44 (63.74, 71.68)	**78.56 (72.2, 83.28)**
	ROUGE	29.36 (24.23, 34.73)	29.08 (24.06, 34.8)	29.92 (25.31, 34.09)	24.86 (19.47, 29.94)	24.81 (16.93, 30.59)	67.90 (63.58, 74.28)	**79.42 (75.15, 84.05)**
	UMLS_Precision	23.02 (17.52, 30.73)	23.32 (18.08, 29.42)	23.14 (18.29, 28.86)	18.28 (13.23, 23.38)	68.87 (64.43, 73.27)	76.09 (71.63, 80.21)	**81.5 (76.81, 86.87)**
	UMLS_Recall	22.71 (17.48, 29.53)	23.74 (18, 30.08)	23.31 (18.29, 27.98)	19.21 (13.38, 24.37)	57.38 (52.49, 63.66)	72.04 (67.59, 76.36)	**74.42 (66.7, 81.19)**
	BERT-Sim	69.42 (66.09, 72.04)	67.7 (64.94, 70.69)	69.14 (66.53, 70.92)	66.93 (63.98, 70.32)	62.35 (61.15, 63.66)	90.93 (89.46, 92.30)	**93.30 (90.99, 95.60)**
Report Generation								
MIMIC-CXR	BLEU	**23.79 (22.62, 24.86)**	0.04 (0.01, 0.08)	11.29 (9.92, 12.86)	22.65 (20.93, 24.06)	11.06 (8.36, 14.43)	20.63 (17.16, 25.43)	19.43 (16.12, 23.25)
	ROUGE	**35.83 (33.7, 37.96)**	2.69 (2.26, 3.15)	13.91 (12.63, 15.29)	27.29 (25.63, 29.04)	15.05 (12.72, 19.54)	25.42 (21.89, 29.47)	26.18 (23.07, 29.86)
	UMLS_Precision	16.75 (15.74, 17.88)	26.67 (11.19, 42.12)	10.50 (8.42, 12.88)	22.36 (20.13, 24.33)	21.80 (19.26, 24.29)	43.64 (36.96, 49.45)	**45.51 (40.47, 52.77)**
	UMLS_Recall	24.93 (22.86, 27.38)	0.52 (0.2, 0.88)	10.71 (8.37, 13.85)	19.64 (17.89, 21.43)	15.97 (12.92, 18.48)	22.73 (19.64, 26.57)	**23.39 (20.18, 27.53)**
	BERT-Sim	65.91 (65.20, 66.70)	34.48 (32.69, 36.02)	49.20 (48.22, 50.35)	66.03 (65.37, 66.83)	63.13 (61.31, 64.87)	64.22 (61.74, 65.97)	**66.77 (64.87, 68.58)**

Respond Letter Table 2: Comparison of proposed RadFM with foundation model baselines on RadBench. The benchmark includes 3 generative-based tasks, medical visual question answering, report generation, and rationale diagnosis. ACC, F1, BLEU, ROUGE, BERT-Sim UMLS_Precision, and UMLS_Recall are reported, and the metrics refer to the average score on all test samples. Numbers within parentheses indicate 95% CI. Percentage (%) signs have been omitted in the table.

Metric	OpenFlamingo [5]	MedVInT [50]	LLaVA-Med [25]	Med-Flamingo [30]	OpenFlamingo [5] (Few-shot)	Med-Flamingo [30] (Few-shot)	RadFM
Medical VQA							
BLEU	6.42 (6.10, 6.67)	1.56 (1.31, 1.97)	21.23 ((20.65, 21.80)	15.55 (14.71, 16.44)	19.93 (18.73, 21.07)	18.68 (17.77, 19.78)	**23.23 (22.16, 24.26)**
ROUGE	28.97 (27.79, 30.12)	3.95 (3.54, 4.51)	25.19 (24.51, 25.95)	20.56 (19.63, 21.66)	26.27 (24.83, 27.55)	24.86 (23.86, 26.15)	**30.88 (29.84, 32.16)**
UMLS_Precision	17.4 (16.2, 18.61)	8.62 (6.80, 10.33)	19.57 (18.84, 20.34)	21.93 (20.48, 23.66)	22.28 (20.42, 24.19)	19.42 (17.75, 21.03)	**22.89 (21.02, 24.48)**
UMLS_Recall	**19.78 (18.54, 20.92)**	1.95 (1.47, 2.46)	18.07 (15.84, 20.36)	12.98 (11.86, 14.11)	17.19 (15.73, 18.62)	14.19 (12.84, 15.55)	17.80 (16.43, 19.08)
BERT-Sim	46.17 (45.55, 46.66)	71.39 (66.94, 75.46)	60.12 (58.83, 61.26)	52.12 (51.37, 52.81)	58.24 (57.59, 58.97)	57.34 (56.64, 58.09)	**72.13 (70.36, 74.23)**
Report Generation							
BLEU	3.25 (2.24, 4.23)	1.73 (1.20,2.30)	8.89 (8.39, 10.37)	9.91 (9.40, 10.37)	1.94 (1.3, 2.71)	4.97 (4.53, 5.40)	**10.21 (9.48, 11.03)**
ROUGE	7.17 (5.26, 9.24)	4.72 (4.21,5.27)	14.21 (13.71, 14.83)	**15.62 (14.96, 16.17)**	3.59 (1.44, 5.47)	6.96 (6.32, 7.44)	15.51 (18.12, 19.98)
UMLS_Precision	1.13 (0.19, 3.3)	9.61 (7.33,11.84)	6.86 (6.62, 7.14)	2.57 (2.14, 3.06)	0.77 (0, 2.26)	2.00 (1.58, 2.51)	**18.97 (18.12, 19.98)**
UMLS_Recall	1.35 (0.19, 3.3)	1.45 (0.95,1.95)	6.00 (5.39, 5.78)	2.03 (1.63, 2.40)	0.71 (0, 2.10)	1.15 (0.87, 1.41)	**9.32 (8.81, 9.89)**
BERT-Sim	37.18 (35.76, 38.45)	38.89 (38.18,39.58)	47.51 (47.12, 47.99)	47.98 (47.6, 48.37)	36.01 (34.42, 37.10)	44.98 (44.26, 45.70)	**56.78 (56.40, 57.22)**
Rationale Diagnosis							
BLEU	4.12 (3.63,4.88)	0.08 (0.02,0.19)	10.82 (10.29, 11.36)	7.65 (7.00,8.37)	18.10 (17.52,18.86)	17.15 (16.4,17.81)	**34.60 (31.69,37.74)**
ROUGE	4.56 (4.10,4.98)	0.67 (0.52,0.83)	14.32 (13.85, 14.82)	7.38 (6.69,7.90)	28.40 (26.88,29.63)	29.02 (27.60,30.28)	**41.89 (39.20,44.77)**
UMLS_Precision	11.58 (8.49,14.90)	7.73 (1.69,15.24)	7.73 (1.69,15.24)	6.01 (5.67, 6.34)	19.26 (18.14,20.51)	21.04 (19.73,22.19)	**42.95 (39.59,46.22)**
UMLS_Recall	0.96 (0.66,1.29)	0.06 (0.01,0.20)	10.22 (8.82, 11.24)	2.17 (1.78,2.66)	16.68 (15.55,17.86)	16.89 (15.71,18.24)	**33.07 (30.93,36.17)**
BERT-Sim	39.20 (38.55,40.02)	29.14 (28.48,29.81)	51.14 (50.90, 51.38)	44.72 (43.97,45.65)	54.11 (53.61,54.57)	54.38 (53.94,54.89)	**68.47 (66.85,70.05)**

Respond Letter Table 3: Comparison of RadFM with SOTA models on disease diagnosis, medical visual question answering, report generation. All models were fine-tuned and evaluated on the same train/test set. AUC, F1, BLEU, and ROUGE are reported, and the metrics refer to the average score on all test samples. For multiple class tasks, the macro-average on classes of the used metrics is adopted. Numbers within parentheses indicate 95% CI.

Dataset	Modality	Metric	SOTA	RadFM
Disease Diagnosis				
VinDr-Mammo	Mammography 2D	AUC	64.5 [44]	**64.76 (64.23, 65.88)**
		F1	N/A	**39.42 (39.37, 39.59)**
CXR14	X-ray 2D	AUC	80.1 [46]	**81.13 (81.07, 81.18)**
		F1	N/A	**30.20 (30.17, 30.22)**
LDCT	CT 3D	AUC	82.1 [44]	**83.23 (81.97, 85.85)**
		F1	N/A	**58.34 (57.38, 61.23)**
MosMedData	CT 3D	AUC	77.47 [31]	**78.33 (76.37, 80.84)**
		F1	50.70	**52.35 (49.26, 55.17)**
COVID-CT	CT 2D	AUC	76.00†	**81.37 (78.00, 82.49)**
		F1	73.35†	**76.11 (74.30, 77.06)**
BraTs2019	MRI 3D	AUC	88.06 [8]	**90.61 (85.66, 92.13)**
		F1	90.36 [8]	**92.21 (91.01, 93.21)**
ADNI	MRI 3D	AUC	79.34 [21]	**80.39 (78.26, 82.44)**
		F1	N/A	**69.88 (68.43, 71.10)**
BTM-17	MRI 2D	AUC	92.80†	**94.47 (92.60, 96.98)**
		F1	70.35†	**74.19 (72.45, 76.31)**
Lung-PET-CT-Dx	PET-CT 3D	AUC	53.44†	**54.57 (51.31, 57.69)**
		F1	36.07†	**37.24 (34.41, 41.53)**
Medical VQA				
MedDiffVQA	X-ray 2D Comparison	Bleu	62.80 [18]	**63.89 (62.27, 64.39)**
		Rogue	N/A	**65.90 (64.48, 63.39)**
		F1	N/A	**59.19 (57.88, 60.43)**
VQA-RAD	Radiology 2D	Bleu	71.03 [7]	**73.44 (66.04, 82.18)**
		Rogue	N/A	**73.81 (67.80, 80.04)**
		F1	N/A	**78.09 (73.54, 81.90)**
SLAKE	Radiology 2D	Bleu	78.6 [42]	**83.16 (79.68, 87.10)**
		Rogue	N/A	**83.65 (80.39, 87.10)**
		F1	78.1 [42]	**84.37 (81.60, 86.78)**
PMC-VQA	Radiology 2D	Bleu	23.69 (20.70, 26.93) [50]	**24.13 (21.01, 27.91)**
		Rogue	**27.20 (24.09, 31.13)** [50]	25.64 (22.73, 29.29)
		F1	43.93 (41.16, 46.43) [50]	**48.50 (46.19, 51.00)**
Report Generation				
IU-Xray	X-ray 2D	Bleu-1	**38.7** [2]	37.88 (35.96, 39.32)
		Bleu-2	24.5 [2]	**24.62 (22.73, 26.94)**
		Bleu-3	16.6 [2]	**17.72 (15.77, 19.69)**
		Bleu-4	**11.1** [2]	10.28 (8.89, 11.64)
		Rogue-L	28.9 [2]	**29.51 (28.09, 30.61)**

† These datasets are not considered a lot as classification tasks. Thus, these scores are obtained by training from scratch with the same architecture as ours to show the effectiveness of RadFM as a pre-trained foundation model.

Respond Letter Table 4: Ablation Studies on a sampled training subset with 0.1 ratios. We analyze how different training facts may affect the final performance, from three aspects: image encoder architectures, patch sizes, and prompt design. "Default" denotes the setting we use in the main text. "LongViT" refers to use the image encoder, introduced in works [13, 43] and "Auto-T" denotes using the autoregressive transformer [14]. The "Patch Size" column denotes changing to larger patch sizes. The "Diversity" denotes using more prompt phrases and "Complexity" denotes to polish the prompts to be more detailed.

Dataset	Metric	Default	Image Encoder		Patch Size		Prompt Design	
			LongViT	Auto-T	64	128	Diversity	Complexity
Disease Diagnosis								
VinDr-Mammo	ACC	53.96 (52.31, 55.32)	53.51 (51.64, 55.18)	54.90 (53.14, 56.65)	53.13 (51.80, 55.69)	52.99 (51.34, 55.82)	53.51 (51.85, 56.35)	53.40 (51.87, 56.19)
	F1	60.33 (58.79, 62.59)	59.55 (57.83, 60.40)	59.89 (58.17, 61.54)	59.62 (58.02, 61.13)	58.64 (57.14, 60.24)	58.54 (56.35, 60.68)	58.21 (56.54, 60.66)
VinDr-SpineXr	ACC	59.67 (56.65, 62.31)	59.64 (56.60, 62.92)	61.19 (57.73, 64.10)	58.41 (55.29, 63.03)	57.41 (54.47, 62.35)	60.19 (57.44, 64.10)	59.30 (56.36, 62.14)
	F1	63.64 (60.60, 66.37)	62.65 (61.49, 68.16)	64.59 (60.75, 67.80)	61.69 (58.79, 65.85)	60.72 (57.96, 64.29)	64.58 (61.39, 68.53)	63.95 (60.93, 66.43)
VinDr-PCXR	ACC	50.77 (45.40, 55.17)	48.47 (43.08, 54.21)	49.13 (44.25, 54.12)	47.03 (42.14, 52.87)	49.43 (45.11, 55.26)	49.13 (44.25, 54.02)	51.05 (45.59, 56.13)
	F1	50.27 (44.69, 55.43)	48.57 (42.44, 54.30)	48.07 (42.88, 53.55)	50.12 (44.46, 57.07)	50.94 (46.78, 56.01)	50.07 (44.26, 55.07)	51.25 (45.29, 56.18)
CXR-Mix	ACC	76.20 (75.65, 76.62)	78.33 (77.07, 78.63)	76.84 (75.60, 77.14)	75.51 (75.16, 75.85)	74.25 (73.87, 74.68)	76.20 (75.65, 76.62)	76.20 (75.65, 76.62)
	F1	76.65 (76.15, 77.17)	78.95 (78.57, 79.26)	76.53 (76.23,77.88)	77.04 (76.67,77.44)	75.64 (75.34,76.01)	76.84 (76.53,77.07)	76.69 (76.24,77.01)
RadChest-CT	ACC	70.09 (69.73, 70.76)	70.25 (69.68, 71.17)	70.59 (69.54, 71.60)	70.31 (69.89, 70.86)	68.42 (69.78, 68.91)	70.52 (69.16, 70.80)	70.21 (69.64, 70.90)
	F1	68.29 (66.75, 68.91)	67.10 (65.52, 68.17)	69.23 (67.41, 70.24)	68.51 (66.05, 69.16)	67.75 (65.15, 68.37)	68.23 (66.31, 68.78)	68.84 (66.84, 68.60)
Medical VQA								
PMC-VQA	BLEU	13.92 (10.66, 16.98)	12.78 (9.88, 15.45)	14.34 (11.06, 17.19)	11.17 (8.04, 14.30)	11.12 (7.95, 14.09)	12.95 (9.56, 15.79)	13.12 (10.12, 15.98)
	ROUGE	14.12 (11.19, 17.36)	14.73 (11.69, 18.27)	15.81 (12.91, 19.38)	13.20 (10.35, 16.57)	12.80 (9.84, 15.95)	14.40 (11.69, 17.80)	14.40 (11.59, 17.774)
	UMLS_Precision	12.45 (9.61, 15.89)	12.71 (9.82, 15.97)	13.62 (10.98, 17.05)	11.64 (8.68, 15.16)	11.53 (8.80, 15.48)	12.03 (9.27, 15.51)	11.84 (9.00, 15.38)
	UMLS_Recall	10.08 (7.87, 13.27)	10.37 (8.06, 14.16)	12.11 (9.86, 16.40)	9.04 (6.71, 12.10)	10.42 (8.15, 13.54)	10.88 (8.29, 13.79)	10.22 (7.92, 13.06)
	BERT-Sim	56.29 (55.33, 60.75)	57.07 (54.95, 59.45)	59.48 (56.37, 61.16)	56.09 (55.23, 58.48)	57.55 (56.68, 60.14)	56.10 (55.04, 58.68)	57.66 (56.82, 60.18)
VQA-RAD	BLEU	36.96 (29.78, 44.20)	37.44 (29.92, 45.88)	38.77 (31.37, 45.38)	37.04 (30.13, 46.00)	34.11 (28.36, 43.05)	38.71 (31.60, 45.53)	38.65 (31.86, 45.86)
	ROUGE	37.96 (31.44,44.75)	38.45 (30.99,45.10)	39.58 (32.74, 36.87)	37.95 (31.72, 44.55)	36.18 (30.47, 43.43)	39.28 (33.57, 46.84)	39.59 (32.75, 46.80)
	UMLS_Precision	60.35 (52.84, 67.06)	62.03 (55.15, 69.18)	62.53 (55.58, 69.35)	61.44 (54.50, 68.04)	59.40 (52.73, 66.45)	61.40 (55.66, 69.35)	61.65 (55.38, 69.55)
	UMLS_Recall	41.52 (34.64, 48.16)	42.48 (35.41, 49.35)	43.25 (36.44, 50.74)	42.58 (35.66, 49.35)	40.81 (33.55, 47.71)	42.68 (36.40, 50.87)	43.28 (36.77, 51.40)
	BERT-Sim	63.66 (61.64, 66.65)	64.87 (62.73, 67.86)	63.47 (62.85, 67.32)	62.82 (60.59, 65.28)	61.67 (60.63, 65.69)	64.56 (62.87, 67.54)	63.27 (61.35, 66.12)
SLAKE	BLEU	48.35 (43.89, 52.59)	49.19 (44.46, 53.45)	51.25 (47.12, 56.19)	46.11 (41.70, 51.31)	44.08 (39.86, 49.30)	50.25 (47.76, 55.44)	48.12 (43.27, 53.59)
	ROUGE	48.89 (44.17, 53.00)	49.75 (44.43, 54.53)	51.84 (46.61, 56.19)	46.60 (41.64, 51.31)	44.61 (39.74, 49.20)	50.84 (46.09, 55.36)	48.78 (44.28, 53.40)
	UMLS_Precision	69.85 (65.03,73.72)	67.53 (63.48,71.59)	71.67 (67.76,75.85)	67.80 (63.65,71.43)	64.98 (63.47,72.01)	68.57 (64.59,72.46)	67.64 (63.20,71.56)
	UMLS_Recall	56.07 (51.74, 62.93)	55.04 (50.94, 62.00)	58.73 (54.54, 67.22)	54.46 (50.37, 62.68)	53.36 (51.17, 62.52)	54.78 (50.55, 62.70)	55.04 (49.87, 62.13)
	BERT-Sim	76.79 (75.89, 77.83)	76.26 (75.04, 77.34)	77.69 (76.62, 78.22)	74.47 (73.19, 75.75)	74.82 (73.65, 75.85)	75.76 (74.48, 77.01)	76.82 (75.80, 78.57)
Report Generation								
MIMIC-CXR	BLEU	12.09 (9.15, 15.69)	11.44 (8.59, 14.27)	10.88 (7.16, 14.83)	12.06 (9.21, 16.34)	11.48 (8.39, 14.52)	13.38 (10.12, 18.95)	10.81 (9.10, 12.96)
	ROUGE	20.30 (17.18, 23.39)	19.93 (17.29, 22.68)	18.12 (15.02, 21.88)	19.98 (17.17, 24.30)	19.36 (17.04, 22.49)	21.50 (17.85, 25.95)	20.50 (18.78, 22.88)
	UMLS_Precision	23.80 (19.26, 24.29)	22.58 (19.26, 24.29)	22.74 (18.87, 24.51)	22.93 (19.48, 24.94)	22.29 (19.25, 24.19)	24.84 (21.89, 27.01)	20.44 (17.81, 22.94)
	UMLS_Recall	17.97 (12.92, 18.48)	15.70 (10.55, 16.32)	13.87 (8.79, 14.72)	15.12 (10.31, 16.77)	15.29 (9.86, 15.83)	16.78 (11.27, 18.04)	14.22 (8.56, 15.35)
	BERT-Sim	63.13 (61.31, 64.87)	61.18 (59.39, 62.88)	61.42 (59.24, 62.34)	61.12 (59.26, 62.68)	62.29 (60.94, 63.63)	63.34 (62.08, 64.36)	60.47 (58.69, 61.32)

Respond Letter Table 5: Description of the collected dataset **Med**ical **M**ultimodal **D**ataset (MedMD) for model pre-training and the filtered dataset **Rad**iology **M**ultimodal **D**ataset (RadMD) for domain-specific fine-tuning. **Filter Strategy** refers the procedure to curate RadMD. **Size** refers to the pair size for both image and image-text data. **Notably**, this table is different from the Extended Table 2 in the main text, *i.e.*, to better show the this table in response letter, some of the unchanged datasets were omitted.

Dataset Name	Description	Filter Strategy	Size	Filter Size
Interleaved Dataset				
PMC-Inline	A dataset containing PMC-papers which links with images through inline reference, *e.g.*, "as shown in Fig. X". This data is generated using the sources from PMC-OA-subset	Filtered out	11M	0
Visual Instruction Tuning Dataset				
Image data				
2D Image-text data				
PMC-OA [27]	A medical dataset contains paired figures and captions collected from PubMed Central.	Filtered out	1.65M	0
PMC-VQA [50]	A medical visual question-answering dataset generated from PMC-OA.	Keep all	413k	413k
PMC-CaseReport	A sub-dataset filtered from PMC-Inline containing 103K cases report papers. We generate VQA pairs by querying ChatGPT. we keep some background information of the patient to form context input. This data is generated using the sources from PMC-OA-subset	Keep radiology	1.1M	438k
MPx-Single	A medical vision-language dataset contains the modality, plane and captions for each image.This data is generated using the sources from MedPix	Keep all	120k	120k
MPx-Multi	A medical vision-language dataset contains findings, discussion, and diagnoses for each case which may contain a series of radiology images. This data is generated using the sources from MedPix	Keep all	39K	39K
2D & 3D Image-text data				
RP3D-Caption	A medical dataset consists of images and corresponding captions. This data is generated using the sources from Radiopaedia	Keep radiology	73k	51k
RP3D-VQA	A medical visual question-answering dataset generated from the captions in RP3D-Caption. This data is generated using the sources from Radiopaedia	Keep radiology	205k	142k
RP3D-Modality	A medical vision-language dataset contains the modality question for each image. This data is generated using the sources from Radiopaedia	Keep radiology	264K	236k
RP3D-Rationale	A medical vision-language dataset contains disease rationale diagnosis for each case which may contain a series of radiology images. This data is generated using the sources from Radiopaedia	Keep radiology	73K	43k

Respond Letter Table 6: Data Process Guideline. We list out how final resolutions are used for each datasets to make the implementation details more clearly. All reolutions sizes are reported in "Height × Width × Depth" format.

Dataset Name	Original Formats and Resolutions	Input Resolutions
RP3D-Caption	3D Radiology JPG Series. Sizes are not unified, averaged as 719 × 706 × 28.	256 × 256 × [4-64][1]
RP3D-VQA	3D Radiology JPG Series. Sizes are not unified, averaged as 719 × 706 × 28.	256 × 256 × [4-64][1]
RP3D-Modality	3D Radiology JPG Series. Sizes are not unified, averaged as 719 × 706 × 28.	256 × 256 × [4-64][1]
RP3D-Rationale	3D Radiology JPG Series. Sizes are not unified, averaged as 719 × 706 × 28.	256 × 256 × [4-64][1]
RadChest-CT [5]	3D Radiology JPG Series. The sizes are unified to 420 × 420 × 402	256 × 256 × 64
MPx-Single	2D Radiology Slices. Sizes are not unified, averaged as 708 × 754.	512 × 512 × 4
MPx-Multi	2D Radiology Slices. Sizes are not unified, averaged as 708 × 754.	512 × 512 × 4
VinDr-Mammo [32]	2D Mammography. Sizes are unified to 1520 × 912 based on RSNA Kaggle Challenge[2]	512 × 512 × 4
VinDr-SpineXR [33]	2D Spine X-ray Images. Size are not unified, totaling averaged as 2542 × 1602.	512 × 512 × 4
VinDr-PCXR [34]	2D Pediatric Chest X-ray Images. Sizes are not unified, around 1643 × 1349 for each image.	512 × 512 × 4
CXR-Mix [45]	2D Chest X-ray Images. Sizes are unified to 512 × 512 × 1 or 224 × 224 based on former papers [45, 12].	512 × 512 × 4.
VQA-RAD [23]	2D Radiology Slices, 256 × 256.	512 × 512 × 4
SLAKE [28]	2D Radiology Slices, 256 × 256.	512 × 512 × 4
PMC-VQA [50]	2D Paper Images. Sizes are not unified, commonly 224 × 224 is used to process the images, like PMC-CLIP [27] and BiomedCLIP [49].	512 × 512 × 4
PMC-CaseReport	2D Paper Images. Sizes are not unified, commonly 224 × 224 is used to process the images, like PMC-CLIP [27] and BiomedCLIP [49].	512 × 512 × 4
PMC-OA [27]	2D Paper Images. Sizes are not unified, commonly 224 × 224 is used to process the images, like PMC-CLIP [27] and BiomedCLIP [49].	512 × 512 × 4
PMC-Inline	2D Paper Images. Sizes are not unified, commonly 224 × 224 is used to process the images, like PMC-CLIP [27] and BiomedCLIP [49].	512 × 512 × 4

[1] Picking a closest multiple of 4 based on original depth, like 1 for 4, 15 for 16.
[2] `https://www.kaggle.com/competitions/rsna-breast-cancer-detection`

Dataset Codes and Files Linking:

Check the following table to see how to process each dataset and how each file in https://huggingface.co/datasets/chaoyi-wu/RadFM_data_csv is linked to each dataset:

Dataset Name	Process Dataset Code	Related Filename
Rad3D-series	jpg2nii Process Code, nii2npy Process Code, Final Datset to Read npy and Related Texts	radiology_article_npy_train/test.json
MPx-series	MedPix Dataset	MedPix_muli_train/test.csv, MedPix_single_train/test.csv
PMC-Inline	Paper-inline Dataset	paper_train.csv (This dataset is not used for evaluation)
PMC-CaseReport	Case-report Dataset	filtered_case_report_train/test.csv
VinDr-Mammo	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	mammo_balance_train/test.csv
VinDr-SpineXR	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	spinexr_balance_train/test.csv
VinDr-PCXR	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	pcxr_balance_train/test.csv
PMC-OA	Pmcoa Dataset	pmcoa_image_caption_train/test.csv
PMC-VQA	vqa Dataset	pmcvaq_train/test.csv
VQA-RAD	vqa Dataset	vqarad_train/test.csv
SLAKE	vqa Dataset	slakevqa_train/test.csv
MIMIC-CXR	CXR Open Captioning Dataset	mimic_caption_train/test.csv
VinDr-CXR	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	chestxray_balance_train_new.csv, chestxray_balance_test.csv
NIH ChestXray14	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	chestxray_balance_train_new.csv, chestxray_balance_test.csv
CheXpert	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	chestxray_balance_train_new.csv, chestxray_balance_test.csv
Covid-CXR2	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	chestxray_balance_train_new.csv, chestxray_balance_test.csv
NLM-TB	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	chestxray_balance_train_new.csv, chestxray_balance_test.csv
Object-CXR	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	chestxray_balance_train_new.csv, chestxray_balance_test.csv
OpenI	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	chestxray_balance_train_new.csv, chestxray_balance_test.csv
RSNA	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	chestxray_balance_train_new.csv, chestxray_balance_test.csv
SIIM-ACR	Diagnosis Open Format Dataset, Diagnosis Close (yes/no) Format Dataset	chestxray_balance_train_new.csv, chestxray_balance_test.csv

Respond Letter Figure 1: The screenshot of the GitHub README file updating.

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Point-point Respond to Referees: NCOMMS-24-22213B

We thank all reviewers for their valuable comments. Here, we aim to resolve the raised concerns point-by-point, and have updated the manuscript accordingly. Please note that, our following indexes of sections, lines, figures, and tables all refer to the revised manuscript by default. All codes, datasets, and models will be released.

Reviewer 1

Q 1.1 *Overall Assessment: The authors have addressed most of our concerns raised during the review process and demonstrated their commitment to improving the paper. Despite the noted limitations, the paper provides valuable contributions to the field of multimodal medical AI. My recommendation would be to accept the paper, with encouragement to incorporate suggestions on dataset enhancement and encoder optimization in future work.*

Reply: Thank you for your constructive suggestions. We will incorporate newer datasets such as CT-RATE, advanced multimodal models like DeepSeek-VL and Qwen2-VL, and enhanced imaging encoders such as CT-ViT and LongViT to strengthen future models. These future directions have been highlighted in the discussion section (**Line 348-365**) as follows:

> While our model significantly outperforms existing open-source multimodal foundation models, we acknowledge that it does not yet meet clinical criteria. Nevertheless, we believe this work provides a starting point for future research and development toward more generalist medical AI models. With the remarkable and rapid advancements in this field, there will be continuous progress in several key areas. For example, more practical and invaluable datasets, such as CT-RATE [7], are expected to be released. Similarly, more powerful general multimodal foundation models, such as the latest DeepSeek-VL [18] and Qwen2-VL [26], will likely emerge, serving as stronger base models along with increasingly sophisticated imaging encoders, like CT-ViT [7] and LongViT [5]. By consistently integrating these distinct impressive advancements in future work, we, together with the entire research community, can drive the evolution of radiology foundation models toward broader adoption in practical clinical applications.. We will release all corresponding data, codes, and models. We believe this can greatly promote the development of medical foundation models.

Q 1.2 *Exclusion of Cases with Missing Metadata: I find the authors' explanation and handling of cases with missing metadata to be reasonable. However, as noted, this approach significantly limits the evaluation for report generation, particularly for 3D data. I strongly suggest exploring the integration of comprehensive datasets like CT-RATE (available: https://huggingface.co/datasets/ibrahimhamamci/CT-RATE) to enhance the robustness of evaluations for 3D medical imaging tasks.*

Reply: Thank you for your valuable feedback. We have also recognized the significance of the recently released CT-Rate dataset. However, as it was released much later than this work, we have investigated it as one of our other ongoing works, namely, *RadGenome-Chest CT* [31] (currently under revision at *Nature Scientific Data*), which we have tried to integrate CT-Rate into our generative VQA training framework.

However, this integration would produce approximately **1.3M** grounded VQA pairs, all based on 3D CT volumes. The high computational cost associated with processing these real thin-slice CT scans in LLM-based frameworks presents a significant challenge, far exceeding the current experimental scale explored in this paper. Moreover, even if CT-Rate were used solely as an evaluation set, we find that the domain gap between CT reports and the training data of current medical foundation models is large, preventing them from generating competitive results without additional fine-tuning. Therefore, we have chosen to leave this direction for future work, as revised in the discussion section referenced in **Q1.1**, and plan to explore it further once sufficient computational resources become available.

Q 1.3 *Model Architecture Details: Despite improvements in reproducibility through updates to the GitHub repository, the manuscript still lacks sufficient detail about the model architecture (not reproducible at all from the paper). While the code is helpful, it would be beneficial for the paper to provide a clearer explanation of the architectural choices and their implications.*

Reply: Thank you very much for your comments. We have revised and polished the related GitHub repository from the following aspects to improve the reproducibility:

- We have added code comments throughout the entire GitHub repository, covering approximately 500 lines, to enhance clarity and understanding for users.
- We have added a new **'A Detailed Code Explanation' Section** in the GitHub README file, offering an explicit guide to help users understand our source code per file and reproduce our work step-by-step.
- We have provided output examples, to help users verify whether the code runs successfully and produces the expected results.

In the main text, we also state a more detailed forward process based on a specific instance in Section 4.2.3.1 Training Details (**Line 651-662**):

> **A Detailed Forward Example.** To better illustrate our model architecture, we present a simple instruction tuning example: a radiology image paired with the text prompt *"Does the case <image> have pneumonia?"*, with the ground truth response *"Yes."* The model forward procedure will include three main steps, *i.e.*. visual encoding, text fusion, and loss calculation. **Visual Encoding:** A 2D image is first expanded into a pseudo-3D format by adding an extra dimension of size 4. It is then processed by a 3D Vision Transformer (ViT) to produce visual tokens. These are compressed to a fixed length of 32 using a perceiver module, ensuring consistent input regardless of image size. **Text Fusion:** The text prompt is tokenized using the LLM's embedding layer, and the *"<image>"* placeholder is replaced with the visual tokens. This fused sequence is input to the LLM's causal self-attention layers for multimodal understanding. **Loss Calculation:** The model predicts the next tokens auto-regressively, and the loss is computed against the ground truth *"Yes"*. During pre-training, the same forward process is used, but the loss is calculated over all text tokens except the image placeholder, following GPT-style training.

We believe these improvements would significantly enhance the reproducibility and usability of our work and can resolve your concerns.

Q 1.4 *Encoder Design: A significant limitation is the continued reliance on their old encoder, which replicates 2D images to artificially create 3D inputs. This approach appears overly simplistic, especially considering that the paper's main contribution is to establish a unified framework for processing both 2D and 3D data. Furthermore, given that the evaluation primarily focuses on 2D data, this replication unnecessarily increases computational costs without substantial benefits. It is worth noting that more advanced vision encoders are available that natively support both 2D and 3D data, providing a more efficient and performance-aligned solution. This issue is not only about performance optimization but also about the practicality of avoiding the computational overhead associated with duplicating 2D images, especially since the majority of the training and test data are inherently 2D.*

Reply: We agree with the reviewer that the encoder design could be more efficient. As demonstrated by the latest multimodal LLMs, such as Qwen2-VL [26], a more suitable approach for processing 2D and 3D data might involve adopting an "any-resolution" design, which leverages the transformer's ability to handle arbitrary input token lengths. This approach could streamline computation and improve efficiency. We appreciate the valuable feedback and treat the architecture update as future work in the discussion section (**Line 311-316**).

Lastly, due to the scale of the dataset (16M image-text pairs), model size (14B parameters), investigating the effects of different components becomes increasingly challenging and prohibitively expensive in terms of both time and computational resources. As future work, we will further break down the problem and investigate each component, ultimately enhancing our understanding and refining the model's performance. This includes, but is not limited to, improved 2D and 3D unified encoding methods, more effective choices for LLM base models, and enhanced training pipelines that address data imbalance arising from data combination.

Reviewer 3

Q 3.1 *Lack of information about data sources and selection criteria. The authors mainly provide this information by citing the primary data sources and listing them in a table. However, this table should summarize what is known about the basic demographic and clinical characteristics of the patients whose images are in the data sets. I realize this information is often missing from the source publications, but some attempt needs to be made to organize and show the information that is available.*

Reply: Thank you for the valuable suggestion. We have gathered all the available demographic and clinical characteristics of the patients for each dataset. However, due to space limitations, we can only present them in a list format in the **Supplementary Section "Detailed Introduction for Different Data Sources"** instead of incorporating them directly into the original table. We hope this will resolve your concerns.

- **RP3D-series** (https://radiopaedia.org/): This dataset leverages cases from Radiopaedia, a comprehensive, wide case collection platform dedicated to radiology. The collected cases encompass a broad spectrum of demographic groups across regions, including adults and children, males and females, and cover diverse radiology-related diseases spanning multiple systems, including Breast, Cardiac, Central Nervous System, Chest, Forensic, Gastrointestinal, Gynecology, Hematology, Head & Neck, Hepatobiliary, Interventional, Musculoskeletal, Obstetrics, Oncology, Pediatrics, Spine, Trauma, Urogenital, and Vascular.
- **MPx-series** (https://medpix.nlm.nih.gov/home): This dataset utilizes cases from MedPix, a comprehensive open-access platform specializing in medical imaging across various regions, sexes, and age groups. Similar to Radiopaedia, MedPix provides extensive coverage of diverse aspects of medical imaging, encompassing over 9,000 topics, with detailed information and statistics accessible at https://medpix.nlm.nih.gov/topiclist.
- **PMC-Inline** (https://huggingface.co/datasets/chaoyi-wu/PMC-Inline): This dataset comprises images extracted from PubMed Central Open Access papers, covering a wide range of clinical conditions, imaging modalities, and demographically diverse patient groups commonly discussed in academic literature. Due to the automated collection process and the variability in reporting formats across publications, detailed patient information and specific clinical characteristics are difficult to track.
- **PMC-CaseReport** (https://huggingface.co/datasets/chaoyi-wu/PMC-CaseReport_original): This dataset leverages the CaseReport subset of PubMed Central Open Access papers. While its demographic and clinical characteristics are comparable to those of PMC-Inline, it focuses on more academically valuable cases that merit detailed discussion in publicly accessible case reports, facilitating clinical knowledge dissemination and education. Similarly, due to the automated collection process and the variability in reporting formats across publications, detailed patient information and specific clinical characteristics are difficult to track.
- **VinDr-Mammo [21]** (https://www.physionet.org/content/vindr-mammo/1.0.0/): The VinDr-Mammo dataset is a large-scale collection of 5,000 full-field digital mammography (FFDM) exams collected from patients in Vietnam between 2018 and 2020. Each exam includes four standard views (MLO and CC for both breasts) and is annotated with BI-RADS categories, breast density levels, and lesion-specific findings such as masses, calcifications, and distortions. The dataset spans a diverse patient demographic with a focus on clinically significant cases (BI-RADS 3–5), and all annotations were independently double-read by expert radiologists.

- **VinDr-SpineXR [22]** (https://www.physionet.org/content/vindr-spinexr/1.0.0/): The VinDr-SpineXR dataset is a large-scale collection of 10,000 radiographs of the spine, collected retrospectively from patients in Vietnam. Each radiograph is accompanied by detailed annotations, including vertebral segmentation and labeling, as well as abnormalities such as fractures, degenerative changes, and other pathologies. The dataset covers a diverse patient demographic and includes both lateral and anteroposterior (AP) views of the spine.
- **VinDr-PCXR [23]** (https://physionet.org/content/vindr-pcxr/1.0.0/): The VinDr-PCXR dataset is a large-scale collection of 18,000 posteroanterior (PA) chest X-ray images, retrospectively collected from patients in Vietnam. The dataset includes detailed image-level annotations for 22 common thoracic findings, such as lung opacities, pleural effusions, and cardiomegaly, as well as normal cases. All annotations were independently double-read by expert radiologists to ensure quality and consistency. The dataset spans a wide range of patient demographics and clinical presentations, reflecting real-world scenarios.
- **RadChest-CT [1]** (https://zenodo.org/record/6406114/): The RadChest-CT dataset is a large-scale collection of chest computed tomography (CT) scans designed to facilitate research and development of AI models for thoracic disease detection and diagnosis. It consists of thousands of high-resolution CT scans encompassing a diverse range of patient demographics and clinical conditions. The dataset includes detailed annotations of lung abnormalities such as nodules, masses, consolidations, pleural effusions, and other thoracic pathologies, as well as normal cases.
- **PMC-OA [16]** (https://huggingface.co/datasets/axiong/pmc_oa_beta): The PMC-OA dataset is a comprehensive collection of full-text biomedical and life sciences research articles sourced from the PubMed Central Open Access (PMC-OA) subset. This dataset includes a wide range of peer-reviewed articles covering various topics, including clinical studies, case reports, reviews, and experimental research. Similar to PMC-Inline, due to the automated collection process and the variability in reporting formats across publications, detailed patient information and specific clinical characteristics are difficult to track.
- **PMC-VQA [32]** (https://huggingface.co/datasets/xmcmic/PMC-VQA): The PMC-VQA dataset is a specialized resource developed for Visual Question Answering (VQA) tasks within the biomedical domain. It is constructed from biomedical figures and their corresponding captions, sourced from the PubMed Central Open Access (PMC-OA) subset. The dataset exhibits demographic and clinical characteristics similar to those of PMC-OA, offering comprehensive coverage across a wide range of biomedical topics and research areas.
- **VQA-RAD [14]** (https://osf.io/89kps/): The VQA-RAD dataset is a benchmark dataset designed for Visual Question Answering (VQA) in the medical imaging domain. It consists of 315 radiology images, including X-rays, CT scans, and MRIs, alongside 3,515 question-answer pairs. The dataset covers a wide range of anatomical regions (e.g., chest, abdomen, and head) and clinical conditions, such as fractures, infections, and tumors. Questions are categorized into two types: closed-ended (e.g., yes/no, multiple-choice) and open-ended (e.g., descriptive answers), reflecting real-world diagnostic queries.
- **SLAKE [17]** (https://www.med-vqa.com/slake/): The SLAKE dataset is designed for Visual Question Answering (VQA) tasks in the medical imaging domain, focusing on clinical reasoning and interpretation. It consists of 642 annotated medical images (e.g., X-rays, CT scans, MRIs) paired with over 14,000 question-answer pairs. These questions are categorized into closed-ended (yes/no or multiple-choice) and open-ended (descriptive answers). SLAKE emphasizes a diverse range of anatomical regions and clinical conditions, including fractures, organ abnormalities, and pathological findings.
- **MIMIC-CXR [13]** (https://physionet.org/content/mimic-cxr/2.0.0): The MIMIC-CXR dataset is a large-scale, publicly available database of chest radiographs and accompanying radiology reports, designed to support research in medical imaging and natural language processing. It contains over 370,000 chest X-ray images from more than 65,000 patients, along with free-text radiology reports that describe clinical findings, impressions, and diagnoses. The dataset encompasses a wide range of patient demographics, clinical conditions (e.g., pneumonia, cardiomegaly, pleural effusion), and imaging views (e.g., frontal, lateral).

- **VinDr-CXR [20]** (https://physionet.org/content/vindr-cxr/1.0.0/): The VinDr-CXR dataset is a curated collection of chest X-ray (CXR) images developed to support research in medical image analysis. It includes over 18,000 chest radiographs from more than 15,000 patients, annotated by experienced radiologists for a wide range of findings, such as lung opacities, cardiomegaly, pleural effusion, and pulmonary nodules. The dataset covers both frontal and lateral views and provides detailed labels at the image and lesion levels, enabling tasks such as disease classification, localization, and segmentation.
- **NIH ChestXray14 [27]** (https://nihcc.app.box.com/v/ChestXray-NIHCC/folder/36938765345): The NIH ChestXray14 dataset is a large-scale dataset of chest X-ray images designed to facilitate research in medical image analysis. It contains over 112,000 chest radiographs from 30,805 unique patients, annotated with labels for 14 common thoracic diseases and findings, such as atelectasis, cardiomegaly, pleural effusion, pneumonia, and pneumothorax. The dataset includes both frontal and lateral views, and the disease labels were generated using natural language processing on the associated radiology reports.NIH ChestXray14 is widely used for tasks like disease classification, anomaly detection, and localization, making it a foundational resource for advancing AI in radiological imaging.
- **CheXpert [11]** (https://physionet.org/content/mimic-cxr/2.0.0): The CheXpert dataset is a large-scale dataset of chest X-rays developed for advancing research in automated chest radiograph interpretation. It contains 224,316 chest radiographs from 65,240 patients, annotated for 14 common thoracic conditions such as pleural effusion, pneumonia, atelectasis, and cardiomegaly. The dataset features both frontal and lateral views, with labels extracted using a rule-based labeler applied to radiology reports, alongside uncertainty labels (e.g., "positive," "negative," or "uncertain") to reflect the inherent ambiguity in medical diagnostics. CheXpert is widely used for tasks like disease classification, uncertainty modeling, and report generation, providing a benchmark for developing robust AI models in medical imaging.
- **Covid-CXR2 [24]** (https://www.kaggle.com/datasets/andyczhao/covidx-cxr2): The COVID-CXR2 dataset is a specialized collection of chest X-ray images curated to support research on COVID-19 detection and related respiratory conditions. It consists of 16,490 positive COVID-19 chest X-ray images from over 2,800 patients, annotated for conditions such as COVID-19 pneumonia, bacterial pneumonia, and healthy cases. The dataset includes both frontal and lateral views, focusing on identifying radiological patterns associated with COVID-19, such as ground-glass opacities and lung consolidation. COVID-CXR2 is widely used for developing AI models aimed at early detection, disease classification, and clinical decision support in the context of pandemic-related diagnostics.
- **NLM-TB [12]** (https://openi.nlm.nih.gov/imgs/collections/NLM-MontgomeryCXRSet.zip & https://openi.nlm.nih.gov/imgs/collections/ChinaSet_AllFiles.zip): The NLM Tuberculosis (TB) datasets, curated by the Communications Engineering Branch at the Lister Hill National Center for Biomedical Communications (LHNCBC), a division of the National Library of Medicine (NLM), are two publicly available collections of de-identified chest X-ray (CXR) images designed to support research in computer-aided screening and diagnosis of pulmonary diseases, with a particular focus on tuberculosis detection. These datasets include the Montgomery County X-ray Set, which contains 138 postero-anterior CXR images acquired from the tuberculosis control program of Montgomery County, Maryland, USA, with 80 normal and 58 abnormal X-rays showing TB manifestations such as effusions and miliary patterns. The images are provided in DICOM format and include radiologist readings as text files. The second dataset, the Shenzhen Hospital X-ray Set, consists of 615 CXR images collected from Shenzhen No. 3 Hospital in Guangdong Province, China, featuring 340 normal and 275 abnormal X-rays with various TB-related abnormalities, available in JPEG format. Both datasets are invaluable for advancing research in automated disease detection and diagnosis, and researchers are encouraged to use them responsibly while adhering to data-sharing restrictions and acknowledging the NLM in any resulting publications.
- **Object-CXR [9]** (https://web.archive.org/web/20201127235812/https://jfhealthcare.github.io/object-CXR/): The Object-CXR dataset is a collection of 10,000 frontal chest X-ray images designed to support research on foreign object detection in medical imaging. The dataset consists of 5,000 chest X-rays containing foreign objects and 5,000 without any foreign objects, collected from approximately 300 township hospitals across China. Annotation efforts were carried out by 12 medically trained radiologists with 1 to 3 years of experience, who manually annotated potential foreign objects located within the lung field. Annotations were made using bounding boxes, bounding ellipses, or masks, depending on the shape

of the objects. Importantly, support devices such as catheters and pacemakers were excluded from annotation. This dataset provides a valuable resource for developing and evaluating object detection models focused on identifying and localizing foreign objects in chest radiographs.

- **OpenI [4]** (https://www.kaggle.com/datasets/raddar/chest-xrays-indiana-university): The Open-i service by the National Library of Medicine (NLM) includes a dedicated collection of 7,470 chest X-rays accompanied by 3,955 radiology reports from the Indiana Network for Patient Care from the hospitals' picture archiving systems. The images and reports were de-identified automatically and then the automatic de-identification was manually verified.
- **RSNA [25]** (https://www.rsna.org/education/ai-resources-and-training/ai-image-challenge/rsna-pneumonia-detection-challenge-2018): The RSNA Pneumonia Detection Challenge 2018 dataset consists of 30,000 frontal chest X-ray images curated from the 112,000-image NIH CXR8 dataset. These include 16,248 posteroanterior views and 13,752 anteroposterior views, along with a test set of 4,527 images. The images, originally in Portable Network Graphics (PNG) format, were converted to Digital Imaging and Communications in Medicine (DICOM) format, with metadata such as patient age, sex, and projection type (AP or PA) added to the DICOM tags. The dataset features detailed manual annotations created by 18 radiologists from 16 institutions, including 12 chest radiologists with an average of 10.6 years of experience. Annotations include whole image labels, 2D region-of-interest (ROI) bounding boxes, and study-level labels for pneumonia detection, created using the md.ai annotation platform. The images underwent pre-processing steps such as standard and histogram normalization, with all protected health information (PHI) removed
- **SIIM-ACR [6]** (https://www.kaggle.com/datasets/jesperdramsch/siim-acr-pneumothorax-segmentation-data): The SIIM-ACR Pneumothorax Segmentation Dataset, created as part of the Society for Imaging Informatics in Medicine (SIIM) and American College of Radiology (ACR) Pneumothorax Segmentation Challenge, consists of a large collection of de-identified chest X-ray images designed to support the development of AI algorithms for detecting and segmenting pneumothorax. The dataset includes thousands of frontal chest radiographs sourced from diverse clinical settings, with annotations provided by expert radiologists. Each image is labeled with pixel-wise segmentation masks (or none, indicating healthy cases) identifying pneumothorax regions, enabling precise localization of the condition.

Q 3.2 *Unclear innovation. I am not certain how the reported work advances the field. A general radiology foundation model is certainly the future of AI in radiology, and an emphasis on data, model development, and external validation seems like a good way to frame the task. A number of commercial entities are working to develop an image-based general radiology foundation model, and there is every reason to believe they are using similar approaches. The manuscript is put forward as a "proof-of-concept", but it's unclear it has addressed any unproven concepts.*

Reply: Thank you for the thoughtful comments. We fully understand your concerns and appreciate the opportunity to clarify. It is important to note that this paper was first released on **August 4, 2023** on *arXiv*, and was submitted to *Nature Medicine* on **September 12, 2023**, following the first turn of external review and approximately six months of revision. It was later transferred to *Nature Communications* on **April 12, 2024**. At that time, the technological approaches to develop such foundation models were far from being established within the broader community, and to our knowledge, this is definitely one of the **first** foundation models for radiology understanding, this can be validated by other published paper.

For example, in the perspective paper *"Multimodal generative AI for medical image interpretation"* published at *Nature* (26 March 2025), our work is recognized as "a push towards more universal vision-language systems", demonstrating that "frameworks leverage the ability of LLMs to generate reports and answer questions on different conditions and modalities without additional labelled data." The paper *"A generalist vision–language foundation model for diverse biomedical tasks"* published in *Nature Medicine* (07 August 2024) cites our work as a critical baseline as "established models." As acknowledged by Reviewer 1 and supported by several other review papers [28, 30, 33, 19, 15], our paper makes "valuable contributions to the field of multimodal medical AI" and provides a clear proof-of-concept framework for the development of radiology foundation models and details the contributions in the revised discussion section (**Line 346-354**):

Specifically, our work demonstrates:

- **Data Collection**: Highlighting the importance of leveraging and combining web-scale radiology data sources to ensure robust and comprehensive model training.
- **Architecture Formulation**: Proposing the unification of all radiology tasks within a generative architecture formulation, carefully designed with task-specific instructions, to develop a versatile and generalist model.
- **Evaluation**: Underlining the need to monitor model performance across a diverse range of radiology tasks to ensure comprehensive validation and broader applicability.

We are also very pleased to see that many recent pre-print works [29, 3, 2, 8, 10] have been inspired by our efforts, validating the impact and influence of our work on this rapidly evolving field as a **proof-of-concept**.

Q 3.3 *Insufficient description of the model. While it is commendable to make the code public, providing it without a high-level explanation it not very helpful. Even with explanation, it is usually challenging to get model code to run due to its complexity. I imagine the average Nature Communications reader does not have the technical expertise to understand how the code works simply by inspecting it. There should be some high level explanation with comparisons to prior art.*

Reply: Thank you for the suggestions. We have revised and polished the related GitHub repository from the following aspects:

- We have added code comments throughout the entire GitHub repository, covering approximately 500 lines, to enhance clarity and understanding for users.
- We have added a new **'A Detailed Code Explanation' Section** in the GitHub README file, offering an explicit guide to help users understand our source code per file and reproduce our work step-by-step.
- We have provided output examples to help users verify whether the code runs successfully and produces the expected results.

In the main text, we also state a more detailed forward process based on a specific instance in Section 4.2.3.1 Training Details (**Line 651-662**):

A Detailed Forward Example. To better illustrate our model architecture, we present a simple instruction tuning example: a radiology image paired with the text prompt *"Does the case <image> have pneumonia?"*, with the ground truth response *"Yes."* The model forward procedure will include three main steps, *i.e.*. visual encoding, text fusion, and loss calculation. **Visual Encoding:** A 2D image is first expanded into a pseudo-3D format by adding an extra dimension of size 4. It is then processed by a 3D Vision Transformer (ViT) to produce visual tokens. These are compressed to a fixed length of 32 using a perceiver module, ensuring consistent input regardless of image size. **Text Fusion:** The text prompt is tokenized using the LLM's embedding layer, and the *"<image>"* placeholder is replaced with the visual tokens. This fused sequence is input to the LLM's causal self-attention layers for multimodal understanding. **Loss Calculation:** The model predicts the next tokens auto-regressively, and the loss is computed against the ground truth *"Yes"*. During pre-training, the same forward process is used, but the loss is calculated over all text tokens except the image placeholder, following GPT-style training.

We believe these improvements will significantly enhance the reproducibility and usability of our work. Additionally, we are delighted to see that, based on our original codebase, many recent works [28, 30, 33, 19, 15] have successfully reproduced our results and conducted comparisons with our methods.

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Point-point Respond to Referees: NCOMMS-24-22213C

We thank Reviewer 3 for his valuable feedback. Here, we aim to resolve the raised concerns point-by-point, and have updated the manuscript accordingly. Please note that our following indexes of sections, lines, figures, and tables all refer to the revised manuscript by default. All codes, datasets, and models will be released.

Reviewer 3

Q 1.1 *Description of data sources. The authors have provided a nice, concise description of each data source but have not attempted to tabulate key characteristics of the sources. While it is common for data sources to omit key descriptive information, I believe it would still be useful to tabulate some basic data characteristics.*

Reply: Thanks a lot for the constructive suggestion. We have added a new table to detail the critical data, as shown in Respond Letter Table 1 and revised accordingly as Supplementary Table 1.

Q 1.2 *Description of innovation. This manuscript is apparently based on an arXiv preprint that was released 20 months ago. At this point, the information is almost historical in nature given AI's current pace of extremely rapid development. The arXiv preprint has been cited 188 times according to Google Scholar, so it has impacted the field. I will leave it to the journal editors to decide whether it is appropriate to publish this work seemingly for historical reasons. At the end of the Discussion section, the authors provide a list of 3 key concepts the work demonstrates. Even 20 months ago, I think these concepts would be viewed as fairly self-evident. The conceptual innovation still is unclear to me. I understand that the author's model is superior to others, but why? Is it simply because they incorporated more data?*

Reply: Thank you very much for recognizing our work as "impacted the field" while we disagree that "20 months ago, these concepts would be viewed as fairly self-evident". We would like to point out that the well-known Nature Perspective "Foundation models for generalist medical artificial intelligence" [1] was published on April 12, 2023, just about four months before the initial release of our paper. At that time, the idea of building "medical foundation models" remains conceptual, with no practical demonstrations proposed. Our work represents one of the first attempts to show the whole community how to practically develop a radiology foundation model, including data curation, training pipeline and evaluation. We present a comprehensive framework for developing radiology foundation models, emphasizing the power of utilizing vast collections of web-scale datasets and training them within a unified generative pipeline, while underscoring the importance of evaluating the models across various tasks to effectively track their overall performance and development progress. Our work demonstrates the technical feasibility and superiority of such newly proposed "medical generalist foundation model" research directions, thus inspiring multiple continued following works. These contributions underscore the novelty of our work, as it goes beyond conceptual discussion to demonstrate concrete methodology-wise "proof-of-concept" that advances the "medical foundation model" field.

Q 1.3 *Model description. The authors have made a reasonable effort to document their code. The high level explanatory section of the README file is a nice addition. I think the authors have done as much as can be expected to make their code reproducible.*

Reply: Thanks a lot for the recognition.

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Respond Letter Table 1 | Summary of used medical imaging datasets. The table highlights key characteristics such as the original purpose of the dataset, whether the population selection was based on clinical criteria, the primary focus of the dataset, the anatomical areas represented, the size of the dataset (number of cases or images), and the country of origin. For 'Original Purpose', 'Education' means for medical education and 'Research' means for machine learning research or biomedical academic research. Clickable hyperlinks to the dataset sources are provided for further details.

Dataset	Original Purpose	Population-Based Selection	Focus of Collection	Anatomy	Imaging Modality	Case/Image Number	Country of Origin
RP3D-series	Educational	No specific criteria	Radiology-related diseases across multiple systems	Various	Various	615K instances	Global
MPx-series	Educational	No specific criteria	Covering wide range of medical imaging topics	Various	Vaious	159K instances	Global
PMC-Inline	Research	Academical cases published in papers	Serving for academical demonstration	Various	Various	11M	Global
PMC-CaseReport	Research	Academic cases published case reports	Serving for academical clinical case demonstration	Various	Various	1.1M	Global
VinDr-Mammo	Research	Patients for breast cancer screening	Mammography with focus on BI-RADS categories	Breast	Digital mammography	5,000 Exams	Vietnam
VinDr-SpineXR	Research	Patients for spine anomalies screening	Spinal lesions detection and classification	Spine	X-ray	10,466 images	Vietnam
VinDr-PCXR	Research	patients younger than ten years and comes with both the localization of critical findings and the classification of common thoracic diseases.	Interpretation of common thoracic diseases	Chest	X-ray	9,125 X-rays	Vietnam
RadChest-CT	Research	Adult patients	Multiple abnormality prediction	Chest	CT	35,747 scans	Not specified
PMC-OA	Research	Academical cases published in papers	Serving for medical imaging pre-training	Various	Various	1.65M Caption Cases	Global
PMC-VQA	Research	Academical cases published in papers	Visual question answering in biomedical domain	Various	Various	413K	Global
VQA-RAD	Research	No specific criteria	Visual question answering in radiology	Chest, Abdomen, Head	Various	315 images with 3,515 VQA pairs	Global
SLAKE	Research	No specific criteria	Medical Visual question answering for clinical reasoning	Various	Various	642 images with 14,000 QA pairs	Global
MIMIC-CXR	Research	Patients with chest X-ray examinations at the Beth Israel Deaconess Medical Center in Boston, MA	Chest X-ray interpretation	Chest	X-rays	377,110 images	USA
VinDr-CXR	Research	Patients from two major hospitals in Vietnam	Chest X-ray interpretation	Chest	X-rays	18,000 images	Vietnam
NIH ChestXray14	Research	f Patients from the year of 1992 to 2015 collected by NIH	14 common diseases	Chest	X-ray	112,000 images	USA
CheXpert	Research	Patient with chest X-ray examinations	14 observations as positive, negative, or uncertain	Chest	X-ray	224,316 images	USA
Covid-CXR2	Research	Patients related to COVID-19	COVID-19 detection	Chest	X-ray	16,490 images	Global
NLM-TB (Montgomery) & NLM-TB (Shenzhen)	Research	Patients related to tuberculosis	Tuberculosis detection	Chest	X-ray	138 (Montgomery), 615 (Shenzhen) images	USA and China
Object-CXR	Research	Patients related to foreign object detection	Foreign object detection	Chest	X-ray	10,000 images	China
OpenI	Educational	No specific criteria	Radiology reports with images	Chest	X-ray	7,470 images	USA
RSNA	Research	Patients related to pneumonia	Pneumonia detection	Chest	X-ray	30,000 images	USA
SIIM-ACR	Research	Patients related to pneumothorax	Pneumothorax detection	Chest	X-ray	12,100 images	Not specified